T3 UV one-loop RGE term comparison

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

Additional general-interaction couplings

Symbol	Field content	Stored coupling / contraction
λ_7	$(H^\dagger)^2(S_1^\dagger)^2$	`lambdaHHdagS1barS1bar`; invariant contraction model-dependent
λ_{12}	$HH^\dagger S_1^\dagger S_2^\dagger$	`lambdaHHdagS1barS2barCross`; invariant contraction model-dependent
λ_8	$(H^\dagger)^2 S_2^2$	`lambdaHHdagS2S2`; invariant contraction model-dependent
λ_{11}	$S_1(S_1^\dagger)^2 S_2^\dagger$	`lambdaS1S1bar2S2bar`; invariant contraction model-dependent
λ_9	$(S_1^\dagger)^2(S_2^\dagger)^2$	`lambdaS1bar2S2bar2`; invariant contraction model-dependent
λ_{10}	$S_1^\dagger S_2 (S_2^\dagger)^2$	`lambdaS1barS2S2bar2`; invariant contraction model-dependent
λ_{X_1}	$(S_2^\dagger)^4$	`lambdaT3ExtraQ000004I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_2}	$S_2\,(S_2^\dagger)^3$	`lambdaT3ExtraQ000013I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_3}	definition unavailable in registry	`lambdaT3ExtraQ000112I1`; symbolic running parameter
λ_{X_4}	definition unavailable in registry	`lambdaT3ExtraQ000202I1`; symbolic running parameter
$\lambda_{X_5}^{(1)}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3ExtraQ000211I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda_{X_5}^{(2)}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3ExtraQ000211I2`; contraction: `(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])`
λ_{X_6}	$(S_1^\dagger)^4$	`lambdaT3ExtraQ000400I1`; contraction: `1`
$\lambda_{X_7}^{(1)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3ExtraQ001102I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda_{X_7}^{(2)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3ExtraQ001102I2`; contraction: `(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])`
λ_{X_8}	definition unavailable in registry	`lambdaT3ExtraQ001201I1`; symbolic running parameter
λ_{X_9}	$S_1\,(S_1^\dagger)^3$	`lambdaT3ExtraQ001300I1`; contraction: `1`
$\lambda_{X_{10}}$	$(H^\dagger)^2\,(S_2^\dagger)^2$	`lambdaT3ExtraQ020002I1`; contraction: `((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund, #2, #3]))`
$\lambda_{X_{11}}$	definition unavailable in registry	`lambdaT3ExtraQ020020I1`; symbolic running parameter
$\lambda_{X_{12}}$	$(H^\dagger)^2\,S_1^\dagger\,S_2^\dagger$	`lambdaT3ExtraQ020101I1`; contraction: `delS2[SU2L, #4, #1, #2]`

$\lambda_{X_{13}}$	definition unavailable in registry	lambdaT3ExtraQ020200I1; symbolic running parameter
$\lambda_{X_{14}}$	$(H^\dagger)^2\, S_1\, S_2$	lambdaT3ExtraQ021010I1; contraction: delS2[SU2L, #4, #1, #2]
$\lambda_{X_{15}}$	$(H^\dagger)^2\, (S_1)^2$	lambdaT3ExtraQ022000I1; contraction: ((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund, #2, #3]))
$\lambda_{X_{16}}$	$H\, H^\dagger\, (S_2^\dagger)^2$	lambdaT3ExtraQ110002I1; contraction: (del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])
$\lambda^{(1)}_{X_{17}}$	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	lambdaT3ExtraQ110101I1; contraction: (del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])
$\lambda^{(2)}_{X_{17}}$	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	lambdaT3ExtraQ110101I2; contraction: (del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])
$\lambda_{X_{18}}$	$H\, H^\dagger\, (S_1^\dagger)^2$	lambdaT3ExtraQ110200I1; contraction: del[SU2L @ fund, #2, #1]
λ_{X_1}	$(S_2^\dagger)^4$	lambdaT3GeneralQ000004I1; contraction: ((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
λ_{X_2}	$S_2\, (S_2^\dagger)^3$	lambdaT3GeneralQ000013I1; contraction: ((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
$\lambda^{(1)}_{X_5}$	$(S_1^\dagger)^2\, S_2\, S_2^\dagger$	lambdaT3GeneralQ000211I1; contraction: del[SU2L @ S2, #3, #4]
$\lambda^{(2)}_{X_5}$	$(S_1^\dagger)^2\, S_2\, S_2^\dagger$	lambdaT3GeneralQ000211I2; contraction: ((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
λ_{X_6}	$(S_1^\dagger)^4$	lambdaT3GeneralQ000400I1; contraction: 1
$\lambda^{(1)}_{X_7}$	$S_1\, S_1^\dagger\, (S_2^\dagger)^2$	lambdaT3GeneralQ001102I1; contraction: del[SU2L @ S2, #3, #4]
$\lambda^{(2)}_{X_7}$	$S_1\, S_1^\dagger\, (S_2^\dagger)^2$	lambdaT3GeneralQ001102I2; contraction: ((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
λ_{X_9}	$S_1\, (S_1^\dagger)^3$	lambdaT3GeneralQ001300I1; contraction: 1
$\lambda_{X_{10}}$	$(H^\dagger)^2\, (S_2^\dagger)^2$	lambdaT3GeneralQ020002I1; contraction: ((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund, #2, #3]))
$\lambda_{X_{12}}$	$(H^\dagger)^2\, S_1^\dagger\, S_2^\dagger$	lambdaT3GeneralQ020101I1; contraction: delS2[SU2L, #4, #1, #2]
$\lambda_{X_{14}}$	$(H^\dagger)^2\, S_1\, S_2$	lambdaT3GeneralQ021010I1; contraction: delS2[SU2L, #4, #1, #2]

$\lambda_{X_{15}}$	$(H^\dagger)^2\,(S_1)^2$	`lambdaT3GeneralQ022000I1`; contraction: `((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund, #2, #3]))`
$\lambda_{X_{16}}$	$H\,H^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ110002I1`; contraction: `(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])`
$\lambda^{(1)}_{X_{17}}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ110101I1`; contraction: `(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])`
$\lambda_{X_{18}}$	$H\,H^\dagger\,(S_1^\dagger)^2$	`lambdaT3GeneralQ110200I1`; contraction: `del[SU2L @ fund, #2, #1]`
$\mu_{\mathrm{T3GeneralS1S2}}$	scalar bilinear	`muT3GeneralS1S2`; see native Lagrangian for contraction
$\mu_{\mathrm{T3GeneralS1Sq}}$	scalar bilinear	`muT3GeneralS1Sq`; see native Lagrangian for contraction
$\mu_{\mathrm{T3GeneralS2Sq}}$	scalar bilinear	`muT3GeneralS2Sq`; see native Lagrangian for contraction
y_1^{conj}	$L\,F\,S_1^\dagger$	`y1Conj`; conjugate-scalar Yukawa
y_2^{conj}	$L\,F\,S_2^\dagger$	`y2Conj`; conjugate-scalar Yukawa

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \to g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

Column	Representative term structure
T1	$\frac{59{g_1}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$\frac{59{g_1}^3}{6}$
A	-2	1	N/A	3	N/A	2	N/A	$\frac{55{g_1}^3}{6}$
A	-4	1	N/A	3	N/A	2	N/A	$\frac{163{g_1}^3}{6}$
B	1	2	N/A	2	N/A	1	N/A	$\frac{25{g_1}^3}{2}$
B	-1	2	N/A	2	N/A	1	N/A	$\frac{43{g_1}^3}{6}$
B	-3	2	N/A	2	N/A	1	N/A	$\frac{25{g_1}^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
C	1	2	N/A	2	N/A	3	N/A	$\frac{41{g_1}^3}{2}$
C	-1	2	N/A	2	N/A	3	N/A	$\frac{43{g_1}^3}{6}$
C	-3	2	N/A	2	N/A	3	N/A	$\frac{41{g_1}^3}{2}$
D	0	3	N/A	1	N/A	2	N/A	$\frac{55{g_1}^3}{6}$
D	2	3	N/A	1	N/A	2	N/A	$\frac{163{g_1}^3}{6}$
D	-2	3	N/A	1	N/A	2	N/A	$\frac{59{g_1}^3}{6}$
E	0	3	N/A	3	N/A	2	N/A	$\frac{59{g_1}^3}{6}$
E	2	3	N/A	3	N/A	2	N/A	$\frac{179{g_1}^3}{6}$
E	-2	3	N/A	3	N/A	2	N/A	$\frac{59{g_1}^3}{6}$
E	-4	3	N/A	3	N/A	2	N/A	$\frac{179{g_1}^3}{6}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{{g_2}^3}{2}$
T2	$-\frac{17{g_2}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	$-\frac{{g_2}^3}{2}$	—
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{{g_2}^3}{2}$	—
A	-4	1	N/A	3	N/A	2	N/A	$-\frac{{g_2}^3}{2}$	—
B	1	2	N/A	2	N/A	1	N/A	—	$-\frac{17{g_2}^3}{6}$
B	-1	2	N/A	2	N/A	1	N/A	—	$-\frac{17{g_2}^3}{6}$
B	-3	2	N/A	2	N/A	1	N/A	—	$-\frac{17{g_2}^3}{6}$
C	1	2	N/A	2	N/A	3	N/A	—	$\frac{31{g_2}^3}{6}$
C	-1	2	N/A	2	N/A	3	N/A	—	$\frac{7{g_2}^3}{6}$
C	-3	2	N/A	2	N/A	3	N/A	—	$\frac{31{g_2}^3}{6}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{{g_2}^3}{2}$	—
D	2	3	N/A	1	N/A	2	N/A	$-\frac{{g_2}^3}{2}$	—
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{{g_2}^3}{2}$	—
E	0	3	N/A	3	N/A	2	N/A	$\frac{{g_2}^3}{6}$	—
E	2	3	N/A	3	N/A	2	N/A	$\frac{{g_2}^3}{6}$	—
E	-2	3	N/A	3	N/A	2	N/A	$\frac{{g_2}^3}{6}$	—
E	-4	3	N/A	3	N/A	2	N/A	$\frac{{g_2}^3}{6}$	—

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7{g_3}^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-7{g_3}^3$
A	-2	1	N/A	3	N/A	2	N/A	$-7{g_3}^3$
A	-4	1	N/A	3	N/A	2	N/A	$-7{g_3}^3$
B	1	2	N/A	2	N/A	1	N/A	$-7{g_3}^3$
B	-1	2	N/A	2	N/A	1	N/A	$-7{g_3}^3$
B	-3	2	N/A	2	N/A	1	N/A	$-7{g_3}^3$
C	1	2	N/A	2	N/A	3	N/A	$-7{g_3}^3$
C	-1	2	N/A	2	N/A	3	N/A	$-7{g_3}^3$
C	-3	2	N/A	2	N/A	3	N/A	$-7{g_3}^3$
D	0	3	N/A	1	N/A	2	N/A	$-7{g_3}^3$
D	2	3	N/A	1	N/A	2	N/A	$-7{g_3}^3$
D	-2	3	N/A	1	N/A	2	N/A	$-7{g_3}^3$
E	0	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$
E	2	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$
E	-2	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$
E	-4	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$

$\beta^{(1)}_{Y_u}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$
T2	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$
T3	$-8{g_3}^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

$\beta^{(1)}_{Y_d}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$
T3	$-8{g_3}^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

$\beta^{(1)}_{Y_e}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta^{(1)}_{Y_e}$$

Terms 1 of 4

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$
T3	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 2\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ Y_e\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$
T3	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 2\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ Y_e\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ Y_e\right)_{ij} + 2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$

Quantity: Y_e

Terms 3 of 4

Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3(\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}))$
T2	$+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3(\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}))$	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	$+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	—
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 2(\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}))$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$

Quantity: Y_e

Terms 4 of 4

Column	Representative term structure
T1	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_1^{\rm conj}\ \left(y_1^{\rm conj}\right)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T2	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(y_2^{\rm conj}\ \left(y_2^{\rm conj}\right)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_1^{\rm conj}\ \left(y_1^{\rm conj}\right)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(y_2^{\rm conj}\ \left(y_2^{\rm conj}\right)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta_{y_1}^{(1)}$

Quantity: y_1

$$16\pi^2\,\mu\frac{dy_1}{d\mu} = \beta_{y_1}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right)(y_1)_{ij}$
T2	$+4\,\mathrm{Tr}\left(y_1\left(y_1^{\mathrm{conj}}\right)^\dagger\right)\left(y_1^{\mathrm{conj}}\right)_{ij}$
T3	$+2T_\nu^{(1)}(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right)(y_1)_{ij}$	$+4\,\mathrm{Tr}\left(y_1\left(y_1^{\mathrm{conj}}\right)^\dagger\right)\left(y_1^{\mathrm{conj}}\right)_{ij}$	$+2T_\nu^{(1)}(y_1)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	$2T_\nu^{(1)}(y_1)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	$2T_\nu^{(1)}(y_1)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+2\,\mathrm{Tr}\left(y_1\left(y_1^{\mathrm{conj}}\right)^\dagger\right)\left(y_1^{\mathrm{conj}}\right)_{ij}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+2\,\mathrm{Tr}\left(y_1\left(y_1^{\mathrm{conj}}\right)^\dagger\right)\left(y_1^{\mathrm{conj}}\right)_{ij}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_1

Terms 2 of 7

Column	Representative term structure
T1	$+\frac{1}{4}(10\left(y_1\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij}+10\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+4\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
T2	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$
T3	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{1}{4}(10\left(y_1\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij}+10\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+4\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$
A	-2	1	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$
A	-4	1	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	—	$-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$
B	-1	2	N/A	2	N/A	1	N/A	—	$-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{4}{g_1}^2\,(y_1)_{ij}$
B	-3	2	N/A	2	N/A	1	N/A	—	$-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	—	$-\frac{33}{4}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$
C	-1	2	N/A	2	N/A	3	N/A	—	$-\frac{33}{4}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{4}{g_1}^2\,(y_1)_{ij}$
C	-3	2	N/A	2	N/A	3	N/A	—	$-\frac{33}{4}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{1}{12}(29\left(y_1\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij}+29\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1\right)_{ij}+6\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+18\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$
D	-2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{1}{12}(29\left(y_1\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij}+29\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1\right)_{ij}+9\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+18\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$
E	-2	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$
E	-4	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$	$-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$

Quantity: y_1

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 3\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 4\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
T2	$+\frac{3}{4}\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}$
T3	$+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 3\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 4\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	$+\frac{3}{4}\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 3\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_1

Terms 4 of 7

Column	Representative term structure
T1	$+\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}$
T2	$T_\nu^{(1)}\ (y_1)_{ij}$
T3	$+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	$+\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}$	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	$+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
B	−1	2	N/A	2	N/A	1	N/A	—	$+T_\nu^{(1)}\ (y_1)_{ij}$	—
B	−3	2	N/A	2	N/A	1	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	$+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+T_\nu^{(1)}\ (y_1)_{ij}$	—
D	2	3	N/A	1	N/A	2	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	$+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
D	−2	3	N/A	1	N/A	2	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	—
E	0	3	N/A	3	N/A	2	N/A	—	$+T_\nu^{(1)}\ (y_1)_{ij}$	—
E	2	3	N/A	3	N/A	2	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$T_\nu^{(1)}\ (y_1)_{ij}$	—

Quantity: y_1

Terms 5 of 7

Column	Representative term structure
T1	$\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$
T2	$+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
T3	$\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$	$+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	$\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_1

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
T2	$\frac{3}{2}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$
T3	$+\frac{1}{4}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	$\frac{3}{2}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$	$+\frac{1}{4}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	—	—

Quantity: y_1

Terms 7 of 7

Column	Representative term structure
T1	$\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)\ (y_1)_{ij}$
T2	$+\frac{1}{2}(\left(y_2\ (y_2)^{\dagger}\ y_1\right)_{ij} + \left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^{\dagger}\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^{\dagger}\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^{\dagger}\ y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)\ (y_1)_{ij}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+\frac{1}{2}(\left(y_2\ (y_2)^{\dagger}\ y_1\right)_{ij} + \left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^{\dagger}\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^{\dagger}\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^{\dagger}\ y_1\right)_{ij})$
E	0	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)\ (y_1)_{ij}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta_{y_2}^{(1)}$

Quantity: y_2

$$16\pi^2\,\mu\frac{dy_2}{d\mu} = \beta_{y_2}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$T_\nu^{(2)}\,(y_2)_{ij}$
T2	$+\frac{1}{2}(\left(y_1\,(y_1)^\dagger\,y_2\right)_{ij} + \left(y_1^{\text{conj}}\,\left(y_1^{\text{conj}}\right)^\dagger\,y_2\right)_{ij} + \left(Y_e\,(Y_e)^\dagger\,y_2\right)_{ij} + 3\left(y_2\,(y_2)^\dagger\,y_2\right)_{ij})$
T3	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	$+\frac{1}{2}(\left(y_1\,(y_1)^\dagger\,y_2\right)_{ij} + \left(y_1^{\text{conj}}\,\left(y_1^{\text{conj}}\right)^\dagger\,y_2\right)_{ij} + \left(Y_e\,(Y_e)^\dagger\,y_2\right)_{ij} + 3\left(y_2\,(y_2)^\dagger\,y_2\right)_{ij})$	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$+T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$+T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	—	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$+T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$T_\nu^{(2)}\,(y_2)_{ij}$	—	$-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$

Quantity: y_2

Terms 2 of 7

Column	Representative term structure
T1	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$
T2	$\mathrm{Tr}\left(y_2^{\mathrm{conj}} \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2\right)_{ij}$
T3	$+2\,\mathrm{Tr}\left(y_2 \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2^{\mathrm{conj}}\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$	$\mathrm{Tr}\left(y_2^{\mathrm{conj}} \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2\right)_{ij}$	$+2\,\mathrm{Tr}\left(y_2 \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2^{\mathrm{conj}}\right)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{15}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—
B	1	2	N/A	2	N/A	1	N/A	$-\frac{15}{4}{g_1}^2 \left(y_2\right)_{ij}$	—	—
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{3}{4}{g_1}^2 \left(y_2\right)_{ij}$	—	—
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{15}{4}{g_1}^2 \left(y_2\right)_{ij}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$-\frac{15}{4}{g_1}^2 \left(y_2\right)_{ij}$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{3}{4}{g_1}^2 \left(y_2\right)_{ij}$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{15}{4}{g_1}^2 \left(y_2\right)_{ij}$	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—
D	2	3	N/A	1	N/A	2	N/A	$-\frac{15}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	$+4\,\mathrm{Tr}\left(y_2 \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2^{\mathrm{conj}}\right)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$-\frac{15}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}{g_1}^2 \left(y_2\right)_{ij}$	$\mathrm{Tr}\left(y_2^{\mathrm{conj}} \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2\right)_{ij}$	$+2\,\mathrm{Tr}\left(y_2 \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \left(y_2^{\mathrm{conj}}\right)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{15}{2}{g_1}^2 \left(y_2\right)_{ij}$	—	—

Quantity: y_2

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 29\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 18\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
T2	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
T3	$\text{Tr}\left(y_2\ (y_1)^\dagger\right)(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+\frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 29\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 18\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—
B	1	2	N/A	2	N/A	1	N/A	—	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	$\text{Tr}\left(y_2\ (y_1)^\dagger\right)(y_1)_{ij}$
B	-3	2	N/A	2	N/A	1	N/A	—	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+\frac{1}{12}(9\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 29\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 18\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_2

Terms 4 of 7

Column	Representative term structure
T1	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
T2	$\frac{3}{2}T_\nu^{(2)}\ (y_2)_{ij}$
T3	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$\frac{3}{2}T_\nu^{(2)}\ (y_2)_{ij}$	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
C	−1	2	N/A	2	N/A	3	N/A	—	$+\frac{3}{2}T_\nu^{(2)}\ (y_2)_{ij}$	—
C	−3	2	N/A	2	N/A	3	N/A	—	$\frac{3}{2}T_\nu^{(2)}\ (y_2)_{ij}$	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 6\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 6\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$

Quantity: y_2

Terms 5 of 7

Column	Representative term structure
T1	$\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)(y_1)_{ij}$
T2	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}+2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}+5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
T3	$2T_\nu^{(2)}\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)(y_1)_{ij}$	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}+2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}+5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$2T_\nu^{(2)}\,(y_2)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$2T_\nu^{(2)}\,(y_2)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$+2T_\nu^{(2)}\,(y_2)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_2

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 3\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 4\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
T2	$+\frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}$
T3	$+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 3\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 4\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$+\frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}$	$+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}$
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 3\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 6\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_2

Terms 7 of 7

Column	Representative term structure
T1	$+\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}$
T2	$2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)(y_2)_{ij}$
T3	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}+2(5\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\right)_{ij}+5\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}+2\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$+\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}$	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)(y_2)_{ij}$	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}+2(5\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\right)_{ij}+5\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}+2\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{M_F}$

Quantity: M_F

$$16\pi^2\,\mu\,\frac{dM_F}{d\mu} = \beta^{(1)}_{M_F}$$

Terms 1 of 6

Column	Representative term structure
T1	$\frac{1}{4}(2(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(\left(y_1^{\text{conj}}\right)^T\ \left(y_1^{\text{conj}}\right)^*\ M_F\right)_{ij}) + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$
T2	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$
T3	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\frac{1}{4}(2(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(\left(y_1^{\text{conj}}\right)^T\ \left(y_1^{\text{conj}}\right)^*\ M_F\right)_{ij}) + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\frac{1}{4}(3(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(\left(y_1^{\text{conj}}\right)^T\ \left(y_1^{\text{conj}}\right)^*\ M_F\right)_{ij}) + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$	$-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$

Quantity: M_F

Terms 2 of 6

Column	Representative term structure
T1	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij} + 3\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$
T2	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$
T3	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij} + 3\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$+\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$	—
D	-2	3	N/A	1	N/A	2	N/A	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 2\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: M_F

Terms 3 of 6

Column	Representative term structure
T1	$+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$
T2	$-6{g_1}^2\,(M_F)_{ij}$
T3	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$	$-6{g_1}^2\,(M_F)_{ij}$	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$
B	-3	2	N/A	2	N/A	1	N/A	$+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$	$-6{g_1}^2\,(M_F)_{ij}$	—
C	1	2	N/A	2	N/A	3	N/A	—	$-6{g_1}^2\,(M_F)_{ij}$	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	$-6{g_1}^2\,(M_F)_{ij}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: M_F

Terms 4 of 6

Column	Representative term structure
T1	$+\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$
T2	$+\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$
T3	$\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$	$+\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	$\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	$\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	$\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	$\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$

Quantity: M_F

Terms 5 of 6

Column	Representative term structure
T1	$-12{g_2}^2\,(M_F)_{ij}$
T2	$\frac{1}{2}(\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij} + \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij} + \left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$
T3	$-6{g_2}^2(\left((M_F)^T\right)_{ij} + (M_F)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$-12{g_2}^2\,(M_F)_{ij}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	$\frac{1}{2}(\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij} + \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij} + \left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$	$-6{g_2}^2(\left((M_F)^T\right)_{ij} + (M_F)_{ij})$
C	-3	2	N/A	2	N/A	3	N/A	$-12{g_2}^2\,(M_F)_{ij}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: M_F

Terms 6 of 6

Column	Representative term structure
T1	$\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(\left(y_1^{\text{conj}}\right)^T\ \left(y_1^{\text{conj}}\right)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$
T2	$\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij} + \left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(\left(y_1^{\text{conj}}\right)^T\ \left(y_1^{\text{conj}}\right)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij} + \left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{m_1^2}$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

Terms 1 of 8

Column	Representative term structure
T1	$m_1^2(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_1}^{(1)})$
T2	$+\mu_{\mathrm{T3GeneralS1Sq}}(8\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 12\,(\lambda_{X_9})^*)$
T3	$-8\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$m_1^2(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_1}^{(1)})$	$+\mu_{\mathrm{T3GeneralS1Sq}}(8\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 12\,(\lambda_{X_9})^*)$	$-8\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	$-8\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	$-8\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
B	1	2	N/A	2	N/A	1	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-6\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-6\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
D	0	3	N/A	1	N/A	2	N/A	—	$+\mu_{\mathrm{T3GeneralS1Sq}}(4\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 20\,(\lambda_{X_9})^*)$	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
E	0	3	N/A	3	N/A	2	N/A	—	$+\mu_{\mathrm{T3GeneralS1Sq}}(4\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 20\,(\lambda_{X_9})^*)$	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-4\,\mathrm{Tr}\left(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\right)$

Quantity: m_1^2

Terms 2 of 8

Column	Representative term structure
T1	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\mathrm{conj}}\right)^T\ \left(y_1^{\mathrm{conj}}\right)^*\right)$
T2	$+8\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$
T3	$+12\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\mathrm{conj}}\right)^T\ \left(y_1^{\mathrm{conj}}\right)^*\right)$	$+8\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$	$+12\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\mathrm{conj}}\right)^T\ \left(y_1^{\mathrm{conj}}\right)^*\right)$	$+4\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$	$+20\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\mathrm{conj}}\right)^T\ \left(y_1^{\mathrm{conj}}\right)^*\right)$	$+4\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$	$+20\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_1^2

Terms 3 of 8

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)} m_2^2$
T2	$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$
T3	$+12\mu_{\mathrm{T3GeneralS2Sq}} \left(\lambda_{X_7}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+6\lambda_{12}^{(1)} m_2^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+6\lambda_{12}^{(1)} m_2^2$	$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	$+12\mu_{\mathrm{T3GeneralS2Sq}} \left(\lambda_{X_7}^{(1)}\right)^*$
A	-4	1	N/A	3	N/A	2	N/A	$+6\lambda_{12}^{(1)} m_2^2$	$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	—
B	1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)} m_2^2$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)} m_2^2$	—	—
B	-3	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)} m_2^2$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)} m_2^2$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)} m_2^2$	—	—
C	-3	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)} m_2^2$	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda_{12}^{(1)} m_2^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+2\lambda_{12}^{(1)} m_2^2$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+2\lambda_{12}^{(1)} m_2^2$	—	$+4\mu_{\mathrm{T3GeneralS2Sq}} \left(\lambda_{X_7}^{(1)}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_1^2

Terms 4 of 8

Column	Representative term structure
T1	$+12\lambda_{X_7}^{(1)}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
T2	$-6{g_1}^2 m_1^2$
T3	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+12\lambda_{X_7}^{(1)}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-6{g_1}^2 m_1^2$	—
A	-4	1	N/A	3	N/A	2	N/A	—	$-24{g_1}^2 m_1^2$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$-6{g_1}^2 m_1^2$	—
D	-2	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_7}^{(1)}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-6{g_1}^2 m_1^2$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	$-6{g_1}^2 m_1^2$	—
E	-2	3	N/A	3	N/A	2	N/A	$+12\lambda_{X_7}^{(1)}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$ $+4\lambda_{X_7}^{(2)}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-6{g_1}^2 m_1^2$	—
E	-4	3	N/A	3	N/A	2	N/A	—	$-24{g_1}^2 m_1^2$	—

Quantity: m_1^2

Terms 5 of 8

Column	Representative term structure
T1	$-\frac{9{g_2}^2 m_1^2}{2}$
T2	$-\frac{3{g_1}^2 m_1^2}{2}$
T3	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$	—
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\right)$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{27{g_1}^2 m_1^2}{2}$	—
C	1	2	N/A	2	N/A	3	N/A	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$	—
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$	$-12\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\right)$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{27{g_1}^2 m_1^2}{2}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_1^2

Terms 6 of 8

Column	Representative term structure
T1	$-8\,\mathrm{Tr}\left(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\right)$
T2	$+6\mu_{\mathrm{T3GeneralS1S2}}\,(\lambda_{11})^*$
T3	$+6\lambda_{11}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-8\,\mathrm{Tr}\left(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\right)$	$+6\mu_{\mathrm{T3GeneralS1S2}}\,(\lambda_{11})^*$	$+6\lambda_{11}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\right)$	$+6\mu_{\mathrm{T3GeneralS1S2}}\,(\lambda_{11})^*$	$+6\lambda_{11}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_1^2

Terms 7 of 8

Column	Representative term structure
T1	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$
T2	$m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$
T3	$-12{g_2}^2 m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12{g_2}^2 m_1^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	0	3	N/A	3	N/A	2	N/A	—	$m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12{g_2}^2 m_1^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$

Quantity: m_1^2

Terms 8 of 8

Column	Representative term structure
T1	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$
T2	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T3	$+4\mu_{\mathrm{T3GeneralS2Sq}}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	—
E	2	3	N/A	3	N/A	2	N/A	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	—
E	-2	3	N/A	3	N/A	2	N/A	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+4\mu_{\mathrm{T3GeneralS2Sq}}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
E	-4	3	N/A	3	N/A	2	N/A	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	—

$\beta^{(1)}_{m_2^2}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

Terms 1 of 8

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
T2	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$
T3	$+4\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_5}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	$+4\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_5}^{(1)}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
A	−4	1	N/A	3	N/A	2	N/A	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
B	1	2	N/A	2	N/A	1	N/A	—	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
C	1	2	N/A	2	N/A	3	N/A	—	$-6\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	$-6\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
D	0	3	N/A	1	N/A	2	N/A	—	$-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	$+12\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_5}^{(1)}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	$-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
E	0	3	N/A	3	N/A	2	N/A	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
E	2	3	N/A	3	N/A	2	N/A	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
E	−4	3	N/A	3	N/A	2	N/A	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$	—

Quantity: m_2^2

Terms 2 of 8

Column	Representative term structure
T1	$+4\lambda_{X_5}^{(1)}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
T2	$-12{g_2}^2 m_2^2$
T3	$-6{g_1}^2 m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_5}^{(1)}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$-12{g_2}^2 m_2^2$	$-6{g_1}^2 m_2^2$
A	-2	1	N/A	3	N/A	2	N/A	—	$-12{g_2}^2 m_2^2$	—
A	-4	1	N/A	3	N/A	2	N/A	—	$-12{g_2}^2 m_2^2$	$-6{g_1}^2 m_2^2$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+12\lambda_{X_5}^{(1)}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	—	$-6{g_1}^2 m_2^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-24{g_1}^2 m_2^2$
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda_{X_5}^{(1)}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$ $+4\lambda_{X_5}^{(2)}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$-12{g_2}^2 m_2^2$	$-6{g_1}^2 m_2^2$
E	2	3	N/A	3	N/A	2	N/A	—	$-12{g_2}^2 m_2^2$	$-24{g_1}^2 m_2^2$
E	-2	3	N/A	3	N/A	2	N/A	—	$-12{g_2}^2 m_2^2$	—
E	-4	3	N/A	3	N/A	2	N/A	—	$-12{g_2}^2 m_2^2$	$-6{g_1}^2 m_2^2$

Quantity: m_2^2

Terms 3 of 8

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)} m_1^2$
T2	$m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
T3	$+\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,(y_2)^\dagger\right) + 20\left(\lambda_{X_2}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)} m_1^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)} m_1^2$	$m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$+\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,(y_2)^\dagger\right) + 20\left(\lambda_{X_2}\right)^*)$
A	-4	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)} m_1^2$	—	—
B	1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)} m_1^2$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)} m_1^2$	—	—
B	-3	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)} m_1^2$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)} m_1^2$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)} m_1^2$	—	—
C	-3	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)} m_1^2$	—	—
D	0	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)} m_1^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)} m_1^2$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)} m_1^2$	—	$+\mu_{\mathrm{T3GeneralS2Sq}}(8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,(y_2)^\dagger\right) + 12\left(\lambda_{X_2}\right)^*)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	$m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$+\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,(y_2)^\dagger\right) + 20\left(\lambda_{X_2}\right)^*)$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 4 of 8

Column	Representative term structure
T1	$-4\,\mathrm{Tr}\left(M_F\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(M_F\right)^\dagger\right)$
T2	$+4\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$+20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(M_F\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(M_F\right)^\dagger\right)$	$+4\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-8\,\mathrm{Tr}\left(M_F\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(M_F\right)^\dagger\right)$	$+8\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+12\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(M_F\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(M_F\right)^\dagger\right)$	$+4\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 5 of 8

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$
T2	$-\frac{9{g_2}^2 m_2^2}{2}$
T3	$-\frac{27{g_1}^2 m_2^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{27{g_1}^2 m_2^2}{2}$
B	-1	2	N/A	2	N/A	1	N/A	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{3{g_1}^2 m_2^2}{2}$
B	-3	2	N/A	2	N/A	1	N/A	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{3{g_1}^2 m_2^2}{2}$
C	1	2	N/A	2	N/A	3	N/A	—	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{27{g_1}^2 m_2^2}{2}$
C	-1	2	N/A	2	N/A	3	N/A	—	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{3{g_1}^2 m_2^2}{2}$
C	-3	2	N/A	2	N/A	3	N/A	—	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{3{g_1}^2 m_2^2}{2}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 6 of 8

Column	Representative term structure
T1	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\right)$
T2	$-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\right)$
T3	$+6\mu_{\mathrm{T3GeneralS1S2}}\ (\lambda_{10})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\right)$	$-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\right)$	$+6\mu_{\mathrm{T3GeneralS1S2}}\ (\lambda_{10})^*$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\right)$	$-12\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\right)$	$+6\mu_{\mathrm{T3GeneralS1S2}}\ (\lambda_{10})^*$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 7 of 8

Column	Representative term structure
T1	$+6\lambda_{10}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$
T2	$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$
T3	$m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+6\lambda_{10}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	—
C	−1	2	N/A	2	N/A	3	N/A	$+6\lambda_{10}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$	$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	—
C	−3	2	N/A	2	N/A	3	N/A	—	$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$
D	2	3	N/A	1	N/A	2	N/A	—	—	$m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 8 of 8

Column	Representative term structure
T1	$m_2^2(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_2}^{(1)})$
T2	$+4\mu_{\mathrm{T3GeneralS1Sq}}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$
T3	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$m_2^2(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_2}^{(1)})$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+4\mu_{\mathrm{T3GeneralS1Sq}}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$

$\beta^{(1)}_{\lambda_1}$

Quantity: λ_1

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
A	−2	1	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	1	2	N/A	2	N/A	1	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	−1	2	N/A	2	N/A	1	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	−3	2	N/A	2	N/A	1	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	1	2	N/A	2	N/A	3	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	−3	2	N/A	2	N/A	3	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	0	3	N/A	1	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	−2	3	N/A	1	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	0	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	−2	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Quantity: λ_1

Terms 2 of 7

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$
T2	$+4\lambda_5\ (\lambda_5)^*$
T3	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{14}}\ (\lambda_{X_{14}})^*$ $+8\lambda_{X_{18}}\ (\lambda_{X_{18}})^*$
A	−2	1	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{12}}\ (\lambda_{X_{12}})^*$ $+24\lambda_{X_{16}}\ (\lambda_{X_{16}})^*$
A	−4	1	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
B	1	2	N/A	2	N/A	1	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{15}}\ (\lambda_{X_{15}})^*$
B	−1	2	N/A	2	N/A	1	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{10}}\ (\lambda_{X_{10}})^*$
C	1	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{15}}\ (\lambda_{X_{15}})^*$
C	−1	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{10}}\ (\lambda_{X_{10}})^*$
D	0	3	N/A	1	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{14}}\ (\lambda_{X_{14}})^*$ $+24\lambda_{X_{18}}\ (\lambda_{X_{18}})^*$
D	2	3	N/A	1	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
D	−2	3	N/A	1	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{12}}\ (\lambda_{X_{12}})^*$ $+8\lambda_{X_{16}}\ (\lambda_{X_{16}})^*$
E	0	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{14}}\ (\lambda_{X_{14}})^*$ $+24\lambda_{X_{18}}\ (\lambda_{X_{18}})^*$
E	2	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
E	−2	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{12}}\ (\lambda_{X_{12}})^*$ $+24\lambda_{X_{16}}\ (\lambda_{X_{16}})^*$
E	−4	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—

Quantity: λ_1

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$
T3	$-9{g_2}^2\lambda_1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$

Quantity: λ_1

Terms 4 of 7

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$+12{\lambda_1}^2$
T3	$+2{\lambda_{HS_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+2{\lambda_{HS_1}^{(1)}}^2$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+2{\lambda_{HS_1}^{(1)}}^2$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+2{\lambda_{HS_1}^{(1)}}^2$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$

Quantity: λ_1

Terms 5 of 7

Column	Representative term structure
T1	$+6{\lambda_{HS_2}^{(1)}}^2$
T2	$+3{\lambda_{HS_2}^{(A)}}^2$
T3	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
A	-2	1	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
A	-4	1	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
B	1	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
C	-1	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
C	-3	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
D	0	3	N/A	1	N/A	2	N/A	$+2{\lambda_{HS_2}^{(1)}}^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+2{\lambda_{HS_2}^{(1)}}^2$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+2{\lambda_{HS_2}^{(1)}}^2$	—	—
E	0	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
E	-2	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
E	-4	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—

Quantity: λ_1

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$
T2	$+8\lambda_7\left(\lambda_7\right)^*$
T3	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_1

Terms 7 of 7

Column	Representative term structure
T1	$+8\lambda_8 \left(\lambda_8\right)^*$
T2	$+4\lambda_{X_{17}}^{(1)} \left(\lambda_{X_{17}}^{(1)}\right)^*$
T3	$+3{\lambda_{HS_1}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+8\lambda_8 \left(\lambda_8\right)^*$	$+4\lambda_{X_{17}}^{(1)} \left(\lambda_{X_{17}}^{(1)}\right)^*$	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+8\lambda_8 \left(\lambda_8\right)^*$	$+4\lambda_{X_{17}}^{(1)} \left(\lambda_{X_{17}}^{(1)}\right)^*$	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
D	-2	3	N/A	1	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

Quantity: $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

Terms 1 of 16

Column	Representative term structure
T1	$\lambda^{(1)}_{S_1}(8T^{(1)}_{\nu} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$
T2	$+\lambda_{X_9}(24\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 108\left(\lambda_{X_9}\right)^{*})$
T3	$+24\left(\lambda_{X_9}\right)^{*}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_1}(8T^{(1)}_{\nu} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$	$+\lambda_{X_9}(24\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 108\left(\lambda_{X_9}\right)^{*})$	$+24\left(\lambda_{X_9}\right)^{*}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 16

Column	Representative term structure
T1	$-16\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$
T2	$-16\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$
T3	$-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-16\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	$-16\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$	$-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	$-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	$-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
B	1	2	N/A	2	N/A	1	N/A	—	—	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 16

Column	Representative term structure
T1	$-8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)$
T2	$+12\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^{*}$
T3	$+288\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)$	$+12\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^{*}$	$+288\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 16

Column	Representative term structure
T1	$+8\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$
T2	$+6{\lambda_{12}^{(1)}}^2$
T3	$+4{\lambda_{HS_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+8\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
A	-2	1	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
A	-4	1	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	1	2	N/A	2	N/A	1	N/A	$+8\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	-1	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	-3	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	1	2	N/A	2	N/A	3	N/A	$+8\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	-1	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	-3	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	0	3	N/A	1	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	-2	3	N/A	1	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 16

Column	Representative term structure
T1	$+10{\lambda_{S_1}^{(1)}}^2$
T2	$8\lambda_{S_1}^{(1)} T_\nu^{(1)}$
T3	$+24\lambda_{X_7}^{(1)} \left(\lambda_{X_7}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+10{\lambda_{S_1}^{(1)}}^2$	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+10{\lambda_{S_1}^{(1)}}^2$	$8\lambda_{S_1}^{(1)} T_\nu^{(1)}$	$+24\lambda_{X_7}^{(1)} \left(\lambda_{X_7}^{(1)}\right)^*$
A	−4	1	N/A	3	N/A	2	N/A	$+10{\lambda_{S_1}^{(1)}}^2$	$8\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
B	1	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	$4\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	$4\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
B	−3	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	$4\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
C	1	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	$6\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	$6\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
C	−3	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	$6\lambda_{S_1}^{(1)} T_\nu^{(1)}$	—
D	0	3	N/A	1	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	$+8\lambda_{X_7}^{(1)} \left(\lambda_{X_7}^{(1)}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
E	−4	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 6 of 16

Column	Representative term structure
T1	$+12{g_1}^4$
T2	$-12{g_1}^2\lambda_{S_1}^{(1)}$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+12{g_1}^4$	$-12{g_1}^2\lambda_{S_1}^{(1)}$	—
A	-4	1	N/A	3	N/A	2	N/A	$+192{g_1}^4$	$-48{g_1}^2\lambda_{S_1}^{(1)}$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	-1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$+12{g_1}^4$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+12{g_1}^4$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$+12{g_1}^4$	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+12{g_1}^4$	—	—
E	-4	3	N/A	3	N/A	2	N/A	$+192{g_1}^4$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 7 of 16

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T2	$-9{g_2}^2\lambda_{S_1}^{(1)}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$
B	−1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$
B	−3	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{243{g_1}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$
C	−1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$
C	−3	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{243{g_1}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—
D	2	3	N/A	1	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—
E	0	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—
E	2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 8 of 16

Column	Representative term structure
T1	$+\frac{\lambda_{12}^{(A)\,2}}{4}$
T2	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$
T3	$+8\lambda_7\,(\lambda_7)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$	$+8\lambda_7\,(\lambda_7)^*$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$	$+8\lambda_7\,(\lambda_7)^*$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)\;\;2}}{4}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 9 of 16

Column	Representative term structure
T1	$+8\lambda_9\,(\lambda_9)^*$
T2	$+24\lambda_{11}\,(\lambda_{11})^*$
T3	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+8\lambda_9\,(\lambda_9)^*$	$+24\lambda_{11}\,(\lambda_{11})^*$	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+8\lambda_9\,(\lambda_9)^*$	$+24\lambda_{11}\,(\lambda_{11})^*$	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_1}^{(A)})$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_1}^{(A)})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 10 of 16

Column	Representative term structure
T1	$+\lambda_{X_9}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)+\frac{352\left(\lambda_{X_9}\right)^*}{3}\right)$
T2	$+\frac{32}{3}\left(\lambda_{X_9}\right)^*\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^\dagger\right)$
T3	$-\frac{248}{45}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\lambda_{X_9}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)+\frac{352\left(\lambda_{X_9}\right)^*}{3}\right)$	$+\frac{32}{3}\left(\lambda_{X_9}\right)^*\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^\dagger\right)$	$-\frac{248}{45}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda_{X_9}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)+\frac{352\left(\lambda_{X_9}\right)^*}{3}\right)$	$+\frac{32}{3}\left(\lambda_{X_9}\right)^*\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^\dagger\right)$	$-\frac{248}{45}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 11 of 16

Column	Representative term structure
T1	$-\frac{248}{45}\,\mathrm{Tr}\Big(y_1\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big)$
T2	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{248}{45}\,\mathrm{Tr}\Big(y_1\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big)$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big)$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{248}{45}\,\mathrm{Tr}\Big(y_1\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big)$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big)$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	—

Quantity: $\lambda^{(1)}_{S_1}$

Terms 12 of 16

Column	Representative term structure
T1	$+\frac{8}{3}\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$
T2	$+\frac{832}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^*$
T3	$+\frac{16}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\frac{8}{3}\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$	$+\frac{832}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^*$	$+\frac{16}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\frac{832}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^*$	$+\frac{16}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 13 of 16

Column	Representative term structure
T1	$+22{g_2}^4$
T2	$+16{\lambda_{S_1}^{(A)}}^2$
T3	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	—
D	2	3	N/A	1	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$
D	−2	3	N/A	1	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$
E	0	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$
E	−2	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$
E	−4	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+16{\lambda_{S_1}^{(A)}}^2$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 14 of 16

Column	Representative term structure
T1	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$
T2	$+\frac{8}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^*+\left(\lambda_{X_5}^{(2)}\right)^*)$
T3	$+\lambda_{X_5}^{(2)}\left(\frac{8\left(\lambda_{X_5}^{(1)}\right)^*}{3}+4\left(\lambda_{X_5}^{(2)}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\frac{8}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^*+\left(\lambda_{X_5}^{(2)}\right)^*)$	$+\lambda_{X_5}^{(2)}\left(\frac{8\left(\lambda_{X_5}^{(1)}\right)^*}{3}+4\left(\lambda_{X_5}^{(2)}\right)^*\right)$
E	2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$	—	—
E	−4	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(32{g_2}^2-48\lambda_{S_1}^{(1)}\right)$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 15 of 16

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$
T3	$+8\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	—
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	—
E	-2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	$+8\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
E	-4	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 16 of 16

Column	Representative term structure
T1	$+\lambda_{X_7}^{(2)}\left(8\left(\lambda_{X_7}^{(1)}\right)^{*}+\frac{20\left(\lambda_{X_7}^{(2)}\right)^{*}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{X_7}^{(2)}\left(8\left(\lambda_{X_7}^{(1)}\right)^{*}+\frac{20\left(\lambda_{X_7}^{(2)}\right)^{*}}{3}\right)$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

Quantity: $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

Terms 1 of 16

Column	Representative term structure
T1	$\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2})$
T2	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$+8\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$+8\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
A	−4	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$+24\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
E	2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
E	−4	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 16

Column	Representative term structure
T1	$+22{g_2}^4$
T2	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(1)}\right)$
T3	$-24{g_2}^2\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(1)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(1)}$
A	−2	1	N/A	3	N/A	2	N/A	$+22{g_2}^4$	—	$-24{g_2}^2\lambda_{S_2}^{(1)}$
A	−4	1	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(1)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{S_2}^{(1)}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{S_2}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{S_2}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{S_2}^{(1)}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{S_2}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{S_2}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(1)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+{g_1}^2\left(32{g_2}^2-48\lambda_{S_2}^{(1)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(1)}$
E	−2	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	—	$-24{g_2}^2\lambda_{S_2}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	$+22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(1)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 16

Column	Representative term structure
T1	$+12{g_1}^4$
T2	$+2{\lambda_{12}^{(1)}}^2$
T3	$+4{\lambda_{HS_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+12{g_1}^4$	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
A	-2	1	N/A	3	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
A	-4	1	N/A	3	N/A	2	N/A	$+12{g_1}^4$	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
B	1	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
B	-1	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
B	-3	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
C	1	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
C	-1	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
C	-3	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
D	0	3	N/A	1	N/A	2	N/A	$+12{g_1}^4$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	$+192{g_1}^4$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
D	-2	3	N/A	1	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	$+12{g_1}^4$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+192{g_1}^4$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	$+12{g_1}^4$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 16

Column	Representative term structure
T1	$+14{\lambda_{S_2}^{(1)}}^2$
T2	$+16{\lambda_{S_2}^{(A)}}^2$
T3	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	—
A	−2	1	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(A)})$
A	−4	1	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	—
B	1	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_2}^{(1)}}^2$	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_2}^{(1)}}^2$	—	—
B	−3	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_2}^{(1)}}^2$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_2}^{(1)}}^2$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_2}^{(1)}}^2$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_2}^{(1)}}^2$	—	—
D	0	3	N/A	1	N/A	2	N/A	$+10{\lambda_{S_2}^{(1)}}^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+10{\lambda_{S_2}^{(1)}}^2$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+10{\lambda_{S_2}^{(1)}}^2$	—	—
E	0	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	—
E	−2	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(A)})$
E	−4	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_2}^{(1)}}^2$	$+16{\lambda_{S_2}^{(A)}}^2$	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 16

Column	Representative term structure
T1	$+\frac{32}{3}\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T2	$+\lambda_{X_2}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)+\frac{352\left(\lambda_{X_2}\right)^{*}}{3}\right)$
T3	$-\frac{248}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{32}{3}\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\lambda_{X_2}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)+\frac{352\left(\lambda_{X_2}\right)^{*}}{3}\right)$	$-\frac{248}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{32}{3}\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\lambda_{X_2}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)+\frac{352\left(\lambda_{X_2}\right)^{*}}{3}\right)$	$-\frac{248}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 6 of 16

Column	Representative term structure
T1	$-\frac{248}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)$
T2	$-\frac{8}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T3	$+\frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{248}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)$	$-\frac{8}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{248}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)$	$-\frac{8}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 7 of 16

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
T2	$+\frac{16}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$
T3	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{8}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$	$+\frac{16}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$+\frac{16}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 8 of 16

Column	Representative term structure
T1	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$
T2	$+\frac{9{g_2}^4}{4}$
T3	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_2}^{(1)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_2}^{(1)}\right)$
B	-1	2	N/A	2	N/A	1	N/A	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
B	-3	2	N/A	2	N/A	1	N/A	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
C	1	2	N/A	2	N/A	3	N/A	$-5\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_2}^{(1)}\right)$
C	-1	2	N/A	2	N/A	3	N/A	$-5\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
C	-3	2	N/A	2	N/A	3	N/A	$-5\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
D	0	3	N/A	1	N/A	2	N/A	$-8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	—
D	2	3	N/A	1	N/A	2	N/A	$-8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 9 of 16

Column	Representative term structure
T1	$+\frac{243{g_1}^4}{4}$
T2	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
T3	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{243{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{243{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 10 of 16

Column	Representative term structure
T1	$+8\lambda_8\,(\lambda_8)^*$
T2	$+8\lambda_9\,(\lambda_9)^*$
T3	$+24\lambda_{10}\,(\lambda_{10})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+8\lambda_8\,(\lambda_8)^*$	$+8\lambda_9\,(\lambda_9)^*$	$+24\lambda_{10}\,(\lambda_{10})^*$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+8\lambda_8\,(\lambda_8)^*$	$+8\lambda_9\,(\lambda_9)^*$	$+24\lambda_{10}\,(\lambda_{10})^*$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 11 of 16

Column	Representative term structure
T1	$+8\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$
T2	$-12{g_1}^2\lambda_{S_2}^{(1)}$
T3	$\lambda_{S_2}^{(1)}(8T_{\nu}^{(2)}+8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	$+8\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	$+8\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$-12{g_1}^2\lambda_{S_2}^{(1)}$	—
D	2	3	N/A	1	N/A	2	N/A	—	$-48{g_1}^2\lambda_{S_2}^{(1)}$	—
D	−2	3	N/A	1	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^{*}$	—	$\lambda_{S_2}^{(1)}(8T_{\nu}^{(2)}+8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 12 of 16

Column	Representative term structure
T1	$+24\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T2	$+\lambda_{X_2}(24\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^{\dagger}\right)+108\left(\lambda_{X_2}\right)^{*})$
T3	$-16\,\mathrm{Tr}\left(y_2\ (y_2)^{\dagger}\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+24\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\lambda_{X_2}(24\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^{\dagger}\right)+108\left(\lambda_{X_2}\right)^{*})$	$-16\,\mathrm{Tr}\left(y_2\ (y_2)^{\dagger}\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 13 of 16

Column	Representative term structure
T1	$-16\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)$
T2	$-8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T3	$+288\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-16\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)$	$-8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+288\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 14 of 16

Column	Representative term structure
T1	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
T2	$+8\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$
T3	$+\lambda_{X_5}^{(2)}\left(8\left(\lambda_{X_5}^{(1)}\right)^* + \frac{20\left(\lambda_{X_5}^{(2)}\right)^*}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+8\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$	$+\lambda_{X_5}^{(2)}\left(8\left(\lambda_{X_5}^{(1)}\right)^* + \frac{20\left(\lambda_{X_5}^{(2)}\right)^*}{3}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 15 of 16

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$
T3	$+\frac{8}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	—
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	—
E	-2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	$+\frac{8}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
E	-4	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+\frac{20{\lambda_{12}^{(\times)}}^2}{3}$	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 16 of 16

Column	Representative term structure
T1	$+\lambda_{X_7}^{(2)}\left(\frac{8\left(\lambda_{X_7}^{(1)}\right)^*}{3}+4\left(\lambda_{X_7}^{(2)}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{X_7}^{(2)}\left(\frac{8\left(\lambda_{X_7}^{(1)}\right)^*}{3}+4\left(\lambda_{X_7}^{(2)}\right)^*\right)$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

Quantity: $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

Terms 1 of 12

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$
T2	$+\lambda_{X_{18}}(8\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(3\left(\lambda_{X_9}\right)^{*} + 4\left(\lambda_{X_{18}}\right)^{*}))$
T3	$+8\left(\lambda_{X_{18}}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$	$+\lambda_{X_{18}}(8\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(3\left(\lambda_{X_9}\right)^{*} + 4\left(\lambda_{X_{18}}\right)^{*}))$	$+8\left(\lambda_{X_{18}}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^{\dagger}\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+\lambda_{X_{18}}(4\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(5\left(\lambda_{X_9}\right)^{*} + 4\left(\lambda_{X_{18}}\right)^{*}))$	$+4\left(\lambda_{X_{18}}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^{\dagger}\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\lambda_{X_{18}}(4\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(5\left(\lambda_{X_9}\right)^{*} + 4\left(\lambda_{X_{18}}\right)^{*}))$	$+4\left(\lambda_{X_{18}}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1\right)^{\dagger}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 2 of 12

Column	Representative term structure
T1	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T2	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+12\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+12\lambda_5\ (\lambda_5)^*$
A	−2	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+12\lambda_5\ (\lambda_5)^*$
A	−4	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+12\lambda_5\ (\lambda_5)^*$
B	1	2	N/A	2	N/A	1	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+6\lambda_5\ (\lambda_5)^*$
B	−1	2	N/A	2	N/A	1	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+6\lambda_5\ (\lambda_5)^*$
B	−3	2	N/A	2	N/A	1	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+6\lambda_5\ (\lambda_5)^*$
C	1	2	N/A	2	N/A	3	N/A	$-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+6\lambda_5\ (\lambda_5)^*$
C	−1	2	N/A	2	N/A	3	N/A	$-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+6\lambda_5\ (\lambda_5)^*$
C	−3	2	N/A	2	N/A	3	N/A	$-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+6\lambda_5\ (\lambda_5)^*$
D	0	3	N/A	1	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$
D	2	3	N/A	1	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+4\lambda_5\ (\lambda_5)^*$
D	−2	3	N/A	1	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+4\lambda_5\ (\lambda_5)^*$
E	0	3	N/A	3	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+4\lambda_5\ (\lambda_5)^*$
E	−2	3	N/A	3	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+4\lambda_5\ (\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	—	$+4\lambda_5\ (\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 12

Column	Representative term structure
T1	$+12\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$
T2	$+12\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T3	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+12\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$	$+12\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
A	−2	1	N/A	3	N/A	2	N/A	—	$+12\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
B	1	2	N/A	2	N/A	1	N/A	—	$+24\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$+24\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+20\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
D	−2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	0	3	N/A	3	N/A	2	N/A	$+20\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	−2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 12

Column	Representative term structure
T1	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$
T2	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3	$+6\lambda_1\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
A	-4	1	N/A	3	N/A	2	N/A	$-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	-4	3	N/A	3	N/A	2	N/A	$-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 5 of 12

Column	Representative term structure
T1	$+4{\lambda^{(1)}_{HS_1}}^{2}$
T2	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$
T3	$\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
A	−2	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
A	−4	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
B	1	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
B	−3	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
C	1	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
C	−3	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
D	0	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
D	2	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	0	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	2	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	−2	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	−4	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 6 of 12

Column	Representative term structure
T1	$+12\lambda_{X_{16}}\left(\lambda^{(1)}_{X_7}\right)^*$
T2	$+12\lambda^{(1)}_{X_7}\left(\lambda_{X_{16}}\right)^*$
T3	$+3{g_1}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+12\lambda_{X_{16}}\left(\lambda^{(1)}_{X_7}\right)^*$	$+12\lambda^{(1)}_{X_7}\left(\lambda_{X_{16}}\right)^*$	$+3{g_1}^4$
A	-4	1	N/A	3	N/A	2	N/A	—	—	$+12{g_1}^4$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	$+3{g_1}^4$
D	-2	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_{16}}\left(\lambda^{(1)}_{X_7}\right)^*$	$+4\lambda^{(1)}_{X_7}\left(\lambda_{X_{16}}\right)^*$	$+3{g_1}^4$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	$+3{g_1}^4$
E	-2	3	N/A	3	N/A	2	N/A	—	$+12\lambda^{(1)}_{X_7}\left(\lambda_{X_{16}}\right)^*$ $+4\lambda^{(2)}_{X_7}\left(\lambda_{X_{16}}\right)^*$	$+3{g_1}^4$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+12{g_1}^4$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 7 of 12

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{9{g_2}^4}{4}$
T3	$-9{g_2}^2\lambda^{(1)}_{HS_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
B	−3	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
C	1	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
C	−1	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
C	−3	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 8 of 12

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$-3{g_1}^2\lambda^{(1)}_{HS_1}$
T3	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{27{g_1}^4}{4}$	$-15{g_1}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{27{g_1}^4}{4}$	$-15{g_1}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^2}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 9 of 12

Column	Representative term structure
T1	$+24\lambda_7\,(\lambda_7)^*$
T2	$+3\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{11})^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+24\lambda_7\,(\lambda_7)^*$	$+3\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{11})^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+24\lambda_7\,(\lambda_7)^*$	$+3\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{11})^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 10 of 12

Column	Representative term structure
T1	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\,(\lambda_{11})^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T2	$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\text{Tr}\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\right) + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\,(\lambda_{11})^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\,(\lambda_{11})^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
C	−3	2	N/A	2	N/A	3	N/A	—	$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\text{Tr}\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\right) + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\text{Tr}\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\right) + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 11 of 12

Column	Representative term structure
T1	$+6{g_2}^4$
T2	$+6{\lambda^{(A)}_{HS_1}}^2$
T3	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	—
D	2	3	N/A	1	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
D	-2	3	N/A	1	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	0	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	-2	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	-4	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_1})$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 12 of 12

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$
T2	$+4\lambda_{X_{16}}(3\left(\lambda^{(1)}_{X_7}\right)^*+\left(\lambda^{(2)}_{X_7}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	—
E	2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	$+4\lambda_{X_{16}}(3\left(\lambda^{(1)}_{X_7}\right)^*+\left(\lambda^{(2)}_{X_7}\right)^*)$
E	−4	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	—

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 12

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$
T2	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+4\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
A	−4	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
B	1	2	N/A	2	N/A	1	N/A	—	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$
B	−1	2	N/A	2	N/A	1	N/A	—	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$
B	−3	2	N/A	2	N/A	1	N/A	—	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$
C	1	2	N/A	2	N/A	3	N/A	—	$-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$
C	−1	2	N/A	2	N/A	3	N/A	—	$-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	$-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$
D	0	3	N/A	1	N/A	2	N/A	—	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+12\lambda_5\,(\lambda_5)^*$
D	2	3	N/A	1	N/A	2	N/A	—	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+12\lambda_5\,(\lambda_5)^*$
D	−2	3	N/A	1	N/A	2	N/A	—	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+12\lambda_5\,(\lambda_5)^*$
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
E	−2	3	N/A	3	N/A	2	N/A	—	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 2 of 12

Column	Representative term structure
T1	$+4\lambda_{X_{18}}\left(\lambda^{(1)}_{X_5}\right)^*$
T2	$+4\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$
T3	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_{18}}\left(\lambda^{(1)}_{X_5}\right)^*$	$+4\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	$+24\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^*$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	$+24\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^*$
D	0	3	N/A	1	N/A	2	N/A	$+12\lambda_{X_{18}}\left(\lambda^{(1)}_{X_5}\right)^*$	$+12\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$	$+12\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	$+12\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	$+12\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$ $+4\lambda^{(2)}_{X_5}\left(\lambda_{X_{18}}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 12

Column	Representative term structure
T1	$+6{g_2}^4$
T2	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$
T3	$+3{g_1}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$
A	−2	1	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	—
A	−4	1	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$-\frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+12{g_1}^4$
D	−2	3	N/A	1	N/A	2	N/A	—	$-\frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	—
E	0	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$
E	2	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+12{g_1}^4$
E	−2	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	—
E	−4	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 12

Column	Representative term structure
T1	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
T2	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T3	$+6\lambda_1\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
B	−1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
C	−1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 5 of 12

Column	Representative term structure
T1	$+4{\lambda^{(1)}_{HS_2}}^2$
T2	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$
T3	$+6{\lambda^{(A)}_{HS_2}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$
A	−2	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$
A	−4	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$
B	1	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
B	−3	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
C	1	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
C	−3	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
D	0	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
D	2	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—
E	0	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$
E	−2	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$
E	−4	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_2}}^2$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 6 of 12

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right) + 4\lambda^{(A)}_{S_2})$
T2	$+\lambda_{X_{16}}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right) + 4(5\left(\lambda_{X_2}\right)^{*} + 4\left(\lambda_{X_{16}}\right)^{*}))$
T3	$+4\left(\lambda_{X_{16}}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right) + 4\lambda^{(A)}_{S_2})$	$+\lambda_{X_{16}}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right) + 4(5\left(\lambda_{X_2}\right)^{*} + 4\left(\lambda_{X_{16}}\right)^{*}))$	$+4\left(\lambda_{X_{16}}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+\lambda_{X_{16}}(8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right) + 4(3\left(\lambda_{X_2}\right)^{*} + 4\left(\lambda_{X_{16}}\right)^{*}))$	$+8\left(\lambda_{X_{16}}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right) + 4\lambda^{(A)}_{S_2})$	$+\lambda_{X_{16}}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right) + 4(5\left(\lambda_{X_2}\right)^{*} + 4\left(\lambda_{X_{16}}\right)^{*}))$	$+4\left(\lambda_{X_{16}}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 7 of 12

Column	Representative term structure
T1	$-2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,Y_e\,\left(Y_e\right)^{\dagger}\right)$
T2	$+20\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$
T3	$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,Y_e\,\left(Y_e\right)^{\dagger}\right)$	$+20\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,Y_e\,\left(Y_e\right)^{\dagger}\right)$	$+12\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\,Y_e\,\left(Y_e\right)^{\dagger}\right)$	$+20\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 8 of 12

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$-9{g_2}^2\lambda_{HS_2}^{(1)}$
T3	$+\frac{27{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{27{g_1}^4}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{27{g_1}^4}{4}$
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 9 of 12

Column	Representative term structure
T1	$-15{g_1}^2\lambda^{(1)}_{HS_2}$
T2	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$
T3	$+3\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-15{g_1}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	$+3\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	-3	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—
C	1	2	N/A	2	N/A	3	N/A	$-15{g_1}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	$+3\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	-3	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 10 of 12

Column	Representative term structure
T1	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{10}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T2	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$+24\lambda_8\left(\lambda_8\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{10}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+24\lambda_8\left(\lambda_8\right)^*$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{10}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+24\lambda_8\left(\lambda_8\right)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 11 of 12

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$\lambda^{(1)}_{HS_2}(4T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$\lambda^{(1)}_{HS_2}(4T^{(2)}_{\nu} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$\lambda^{(1)}_{HS_2}(4T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
D	2	3	N/A	1	N/A	2	N/A	—	$\lambda^{(1)}_{HS_2}(4T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_2}(4T^{(2)}_{\nu} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 12 of 12

Column	Representative term structure
T1	$+4\lambda_{X_{18}}(3\left(\lambda^{(1)}_{X_5}\right)^* + \left(\lambda^{(2)}_{X_5}\right)^*)$
T2	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_{18}}(3\left(\lambda^{(1)}_{X_5}\right)^* + \left(\lambda^{(2)}_{X_5}\right)^*)$	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$
E	2	3	N/A	3	N/A	2	N/A	—	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$
E	−2	3	N/A	3	N/A	2	N/A	—	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$
E	−4	3	N/A	3	N/A	2	N/A	—	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

Quantity: $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

Terms 1 of 17

Column	Representative term structure
T1	$\lambda^{(1)}_{12}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 2T^{(2)}_{\nu} + 4\lambda^{(A)}_{S_2})$
T2	$+\lambda^{(1)}_{X_5}(8\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(4\left(\lambda^{(1)}_{X_5}\right)^{*} + 3\left(\lambda_{X_9}\right)^{*}))$
T3	$+8\left(\lambda^{(1)}_{X_5}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{12}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 2T^{(2)}_{\nu} + 4\lambda^{(A)}_{S_2})$	$+\lambda^{(1)}_{X_5}(8\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(4\left(\lambda^{(1)}_{X_5}\right)^{*} + 3\,(\lambda_{X_9})^{*}))$	$+8\left(\lambda^{(1)}_{X_5}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+\lambda^{(1)}_{X_5}(4\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(4\left(\lambda^{(1)}_{X_5}\right)^{*} + 5\,(\lambda_{X_9})^{*}))$	$+4\left(\lambda^{(1)}_{X_5}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\lambda^{(1)}_{X_5}(4\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 4(4\left(\lambda^{(1)}_{X_5}\right)^{*} + 5\,(\lambda_{X_9})^{*}))$	$+4\left(\lambda^{(1)}_{X_5}\right)^{*}\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 17

Column	Representative term structure
T1	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$
T2	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_2\ (y_2)^\dagger\right)$
T3	$+4\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
A	−2	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+4\lambda_5\,(\lambda_5)^*$
A	−4	1	N/A	3	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+4\lambda_5\,(\lambda_5)^*$
B	1	2	N/A	2	N/A	1	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+3\lambda_5\,(\lambda_5)^*$
B	−1	2	N/A	2	N/A	1	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+3\lambda_5\,(\lambda_5)^*$
B	−3	2	N/A	2	N/A	1	N/A	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+3\lambda_5\,(\lambda_5)^*$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+3\lambda_5\,(\lambda_5)^*$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$+3\lambda_5\,(\lambda_5)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$+3\lambda_5\,(\lambda_5)^*$
D	0	3	N/A	1	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_2\ (y_2)^\dagger\right)$	$+4\lambda_5\,(\lambda_5)^*$
D	2	3	N/A	1	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+4\lambda_5\,(\lambda_5)^*$
D	−2	3	N/A	1	N/A	2	N/A	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	$+4\lambda_5\,(\lambda_5)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	$+2\lambda_5\,(\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+2\lambda_5\,(\lambda_5)^*$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+2\lambda_5\,(\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+2\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 17

Column	Representative term structure
T1	$+12\lambda_{X_9}\left(\lambda_{X_5}^{(1)}\right)^*$
T2	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T3	$-12{g_2}^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+12\lambda_{X_9}\left(\lambda_{X_5}^{(1)}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-12{g_2}^2\lambda_{12}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-12{g_2}^2\lambda_{12}^{(1)}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2\lambda_{12}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{12}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{12}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{12}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{12}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{12}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{12}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$+20\lambda_{X_9}\left(\lambda_{X_5}^{(1)}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-12{g_2}^2\lambda_{12}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-12{g_2}^2\lambda_{12}^{(1)}$
D	-2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-12{g_2}^2\lambda_{12}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	—	$+2\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-24{g_2}^2\lambda_{12}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-24{g_2}^2\lambda_{12}^{(1)}$
E	-2	3	N/A	3	N/A	2	N/A	—	$+2\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-24{g_2}^2\lambda_{12}^{(1)}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$-24{g_2}^2\lambda_{12}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 17

Column	Representative term structure
T1	$-6{g_1}^2\lambda_{12}^{(1)}$
T2	$+4{\lambda_{12}^{(1)}}^2$
T3	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
A	-4	1	N/A	3	N/A	2	N/A	$-30{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	$-15{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	$-15{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$-15{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	$-15{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$-6{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	$-30{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
D	-2	3	N/A	1	N/A	2	N/A	$-6{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$-30{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	—
E	-2	3	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	—
E	-4	3	N/A	3	N/A	2	N/A	$-30{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 17

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3	$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
A	−2	1	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_2}^{(A)})$
A	−4	1	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
B	1	2	N/A	2	N/A	1	N/A	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
B	−3	2	N/A	2	N/A	1	N/A	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
C	1	2	N/A	2	N/A	3	N/A	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
C	−3	2	N/A	2	N/A	3	N/A	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
D	2	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	0	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 17

Column	Representative term structure
T1	$+\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 4(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*))$
T2	$+4\left(\lambda_{X_7}^{(1)}\right)^*\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 4(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*))$	$+4\left(\lambda_{X_7}^{(1)}\right)^*\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+\lambda_{X_7}^{(1)}(8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 4(3\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*))$	$+8\left(\lambda_{X_7}^{(1)}\right)^*\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 4(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*))$	$+4\left(\lambda_{X_7}^{(1)}\right)^*\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 7 of 17

Column	Representative term structure
T1	$+20\lambda_{X_2}\left(\lambda_{X_7}^{(1)}\right)^*$
T2	$\lambda_{12}^{(1)}(4T_\nu^{(1)}+2T_\nu^{(2)}+4\lambda_{S_2}^{(A)})$
T3	$+48{g_1}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+20\lambda_{X_2}\left(\lambda_{X_7}^{(1)}\right)^*$	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	$\lambda_{12}^{(1)}(4T_\nu^{(1)}+2T_\nu^{(2)}+4\lambda_{S_2}^{(A)})$	$+48{g_1}^4$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	$+48{g_1}^4$
D	−2	3	N/A	1	N/A	2	N/A	$+12\lambda_{X_2}\left(\lambda_{X_7}^{(1)}\right)^*$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	$+48{g_1}^4$
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+48{g_1}^4$

Quantity: $\lambda_{12}^{(1)}$

Terms 8 of 17

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$+\frac{9{g_2}^4}{4}$
T3	$+\frac{27{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9{g_2}^4}{4}$	$+\frac{27{g_1}^4}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9{g_2}^4}{4}$	$+\frac{3{g_1}^4}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9{g_2}^4}{4}$	$+\frac{27{g_1}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$+\frac{27{g_1}^4}{4}$
C	-1	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$+\frac{3{g_1}^4}{4}$
C	-3	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$+\frac{27{g_1}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\frac{63{g_2}^4}{5}$	—
E	2	3	N/A	3	N/A	2	N/A	—	$+\frac{63{g_2}^4}{5}$	—
E	-2	3	N/A	3	N/A	2	N/A	—	$+\frac{63{g_2}^4}{5}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	$+\frac{63{g_2}^4}{5}$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 9 of 17

Column	Representative term structure
T1	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$
T2	$-4\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\ y_2\ {(y_1)}^\dagger\right)$
T3	$+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	$-4\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\ y_2\ {(y_1)}^\dagger\right)$	$+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	$-3\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\ y_2\ {(y_1)}^\dagger\right)$	$+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 10 of 17

Column	Representative term structure
T1	$+\lambda_{X_{17}}^{(1)}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$
T2	$+24\lambda_9\,(\lambda_9)^*$
T3	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda_{X_{17}}^{(1)}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$	$+24\lambda_9\,(\lambda_9)^*$	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$ $+9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda_{X_{17}}^{(1)}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$	$+24\lambda_9\,(\lambda_9)^*$	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$ $+9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 11 of 17

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$
T2	$-\frac{9}{2}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	$-\frac{9}{2}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	$-\frac{9}{2}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	$-\frac{9}{2}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 12 of 17

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$
T2	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_1}^{(A)})$
T3	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_1}^{(A)})$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 13 of 17

Column	Representative term structure
T1	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_2\,\left(y_2\right)^{\dagger}\right)$
T2	$+4\lambda_{X_9}(5\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$
T3	$+4\lambda_{X_5}^{(2)}(\left(\lambda_{X_5}^{(2)}\right)^{*}+\left(\lambda_{X_9}\right)^{*})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_2\,\left(y_2\right)^{\dagger}\right)$	$+4\lambda_{X_9}(5\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$	$+4\lambda_{X_5}^{(2)}(\left(\lambda_{X_5}^{(2)}\right)^{*}+\left(\lambda_{X_9}\right)^{*})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 14 of 17

Column	Representative term structure
T1	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T2	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T3	$+16{\lambda_{12}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+16{\lambda_{12}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+16{\lambda_{12}^{(A)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+16{\lambda_{12}^{(A)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$+16{\lambda_{12}^{(A)}}^2$

Quantity: $\lambda_{12}^{(1)}$

Terms 15 of 17

Column	Representative term structure
T1	$+4\lambda_{12}^{(\times)\,2}$
T2	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T3	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)\,2}$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)\,2}$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)\,2}$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)\,2}$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 16 of 17

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$
T2	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$
T3	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 17 of 17

Column	Representative term structure
T1	$+4\lambda_{X_2}(5\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
T2	$+4\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + \left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_2}(5\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$	$+4\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_5}$

Quantity: λ_5

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

Terms 1 of 10

Column	Representative term structure
T1	$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right) + 4\lambda_{HS_2}^{(A)})$
T2	$+4\left(\lambda_{X_{14}}\right)^*\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$
T3	$+4\left(\lambda_{X_{14}}\right)^*(\left(\lambda_{X_5}^{(1)}\right)^* + 2\left(\lambda_{X_{18}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right) + 4\lambda_{HS_2}^{(A)})$	$+4\left(\lambda_{X_{14}}\right)^*\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	$+4\left(\lambda_{X_{14}}\right)^*(\left(\lambda_{X_5}^{(1)}\right)^* + 2\left(\lambda_{X_{18}}\right)^*)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+2\left(\lambda_{X_{14}}\right)^*\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	$+4\left(\lambda_{X_{14}}\right)^*(\left(\lambda_{X_5}^{(1)}\right)^* + 2\left(\lambda_{X_{18}}\right)^*)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+2\left(\lambda_{X_{14}}\right)^*\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 2 of 10

Column	Representative term structure
T1	$-\frac{21{g_2}^2\lambda_5}{2}$
T2	$-\frac{9{g_1}^2\lambda_5}{2}$
T3	$+2\lambda_{12}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
A	-4	1	N/A	3	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
B	1	2	N/A	2	N/A	1	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
C	-1	2	N/A	2	N/A	3	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
C	-3	2	N/A	2	N/A	3	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_5}{2}$	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$
E	-4	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_5}{2}$	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 10

Column	Representative term structure
T1	$+2\lambda_1\lambda_5$
T2	$+4\lambda^{(1)}_{HS_1}\lambda_5$
T3	$+4\lambda^{(1)}_{HS_2}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
A	-2	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
A	-4	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
B	-1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
B	-3	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
C	-1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
C	-3	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
D	2	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
D	-2	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
E	2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
E	-2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$
E	-4	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$

Quantity: λ_5

Terms 4 of 10

Column	Representative term structure
T1	$\lambda_5 (2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda_{HS_2}^{(A)})$
T2	$+2\left(\lambda_{X_{12}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$+4\lambda_{X_7}^{(1)}\left(\lambda_{X_{12}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda_{HS_2}^{(A)})$	$+2\left(\lambda_{X_{12}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+4\lambda_{X_7}^{(1)}\left(\lambda_{X_{12}}\right)^*$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+4\left(\lambda_{X_{12}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+4\lambda_{X_7}^{(1)}\left(\lambda_{X_{12}}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	$+2\left(\lambda_{X_{12}}\right)^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+4\lambda_{X_7}^{(1)}\left(\lambda_{X_{12}}\right)^*$ $-2\lambda_{X_7}^{(2)}\left(\lambda_{X_{12}}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 5 of 10

Column	Representative term structure
T1	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$
T2	$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda_{HS_2}^{(A)})$
T3	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda_{HS_2}^{(A)})$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 6 of 10

Column	Representative term structure
T1	$-9{g_2}^2\lambda_5$
T2	$-9{g_1}^2\lambda_5$
T3	$-\frac{2}{3}\left(\lambda_7\right)^* \mathrm{Tr}\left(y_2\ \left(y_1\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_5$	$-9{g_1}^2\lambda_5$	—
B	-1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_5$	$-3{g_1}^2\lambda_5$	$-\frac{2}{3}\left(\lambda_7\right)^* \mathrm{Tr}\left(y_2\ \left(y_1\right)^\dagger\right)$
B	-3	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_5$	$-9{g_1}^2\lambda_5$	—
C	1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_5$	$-9{g_1}^2\lambda_5$	—
C	-1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_5$	$-3{g_1}^2\lambda_5$	—
C	-3	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_5$	$-9{g_1}^2\lambda_5$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 7 of 10

Column	Representative term structure
T1	$+2\,(\lambda_8)^*\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\Big)$
T2	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3	$-\,(\lambda_7)^*\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+2\,(\lambda_8)^*\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\Big)$	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	—
C	-1	2	N/A	2	N/A	3	N/A	$+3\,(\lambda_8)^*\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\Big)$	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-\,(\lambda_7)^*\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
C	-3	2	N/A	2	N/A	3	N/A	—	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 8 of 10

Column	Representative term structure
T1	$\lambda_5 (T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$
T2	$\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$
T3	$\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda_5 (T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	$\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 9 of 10

Column	Representative term structure
T1	$\lambda_5 \left(T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}} \left(y_1^{\mathrm{conj}} \right)^\dagger \right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$
T2	$+2 \left(\lambda_{X_{14}} \right)^* (2 \left(\lambda_{X_5}^{(1)} \right)^* - \left(\lambda_{X_5}^{(2)} \right)^* + 4 \left(\lambda_{X_{18}} \right)^*)$
T3	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda_5 \left(T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}} \left(y_1^{\mathrm{conj}} \right)^\dagger \right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$	$+2 \left(\lambda_{X_{14}} \right)^* (2 \left(\lambda_{X_5}^{(1)} \right)^* - \left(\lambda_{X_5}^{(2)} \right)^* + 4 \left(\lambda_{X_{18}} \right)^*)$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$

Quantity: λ_5

Terms 10 of 10

Column	Representative term structure
T1	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}} \left(y_2^{\mathrm{conj}} \right)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_d \left(Y_d \right)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e \left(Y_e \right)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u \left(Y_u \right)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}} \left(y_2^{\mathrm{conj}} \right)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_d \left(Y_d \right)^\dagger \right) + 2\,\mathrm{Tr}\left(Y_e \left(Y_e \right)^\dagger \right) + 6\,\mathrm{Tr}\left(Y_u \left(Y_u \right)^\dagger \right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$
E	-4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$
T3	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
B	−1	2	N/A	2	N/A	1	N/A	$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
B	−3	2	N/A	2	N/A	1	N/A	$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
C	1	2	N/A	2	N/A	3	N/A	—	$-\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
C	−1	2	N/A	2	N/A	3	N/A	—	$-\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	$-\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	—	$+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$
E	−2	3	N/A	3	N/A	2	N/A	—	$+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	$+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 9

Column	Representative term structure
T1	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{12}^{(A)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
B	-1	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
B	-3	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
C	-1	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
C	-3	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(16{g_2}^2-30\lambda_{12}^{(A)}\right)$	$-24{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(16{g_2}^2-30\lambda_{12}^{(A)}\right)$	$-24{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 9

Column	Representative term structure
T1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
B	−1	2	N/A	2	N/A	1	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
B	−3	2	N/A	2	N/A	1	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
C	−1	2	N/A	2	N/A	3	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
C	−3	2	N/A	2	N/A	3	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
E	2	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
E	−4	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 9

Column	Representative term structure
T1	$+4\lambda_{X_{17}}^{(1)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{X_{17}}^{(1)}\right)^*$
T3	$-64\lambda_9\left(\lambda_9\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+4\lambda_{X_{17}}^{(1)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{X_{17}}^{(1)}\right)^*$	$-64\lambda_9\left(\lambda_9\right)^*$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+4\lambda_{X_{17}}^{(1)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{X_{17}}^{(1)}\right)^*$	$-64\lambda_9\left(\lambda_9\right)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 5 of 9

Column	Representative term structure
T1	$-4\lambda_{10}(5\,(\lambda_{10})^{*}+(\lambda_{11})^{*})$
T2	$-4\lambda_{11}((\lambda_{10})^{*}+5\,(\lambda_{11})^{*})$
T3	$+{g_1}^{2}\left(-6{g_2}^{2}-3\lambda_{12}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-4\lambda_{10}(5\,(\lambda_{10})^{*}+(\lambda_{11})^{*})$	$-4\lambda_{11}((\lambda_{10})^{*}+5\,(\lambda_{11})^{*})$	$+{g_1}^{2}\left(-6{g_2}^{2}-3\lambda_{12}^{(A)}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-4\lambda_{10}(5\,(\lambda_{10})^{*}+(\lambda_{11})^{*})$	$-4\lambda_{11}((\lambda_{10})^{*}+5\,(\lambda_{11})^{*})$	$+{g_1}^{2}\left(-6{g_2}^{2}-3\lambda_{12}^{(A)}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 6 of 9

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$
T2	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$
T3	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 7 of 9

Column	Representative term structure
T1	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_2\,\left(y_2\right)^{\dagger}\right)$
T2	$-\frac{1}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^{*}$
T3	$-6{g_1}^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_2\,\left(y_2\right)^{\dagger}\right)$	$-\frac{1}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^{*}$	$-6{g_1}^2\lambda_{12}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	$-\frac{1}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^{*}$	$-6{g_1}^2\lambda_{12}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 8 of 9

Column	Representative term structure
T1	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T2	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$
T3	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 9 of 9

Column	Representative term structure
T1	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T2	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	—	—

$\beta^{(1)}_{\lambda_{12}^{(\times)}}$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(\times)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(\times)}}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)})$
T2	$+\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\left(\lambda_{X_5}^{(1)}\right)^* + 10\left(\lambda_{X_5}^{(2)}\right)^* + 4\left(\lambda_{X_9}\right)^*)$
T3	$+2\left(\lambda_{X_5}^{(2)}\right)^*\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)})$	$+\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 8\left(\lambda_{X_5}^{(1)}\right)^* + 10\left(\lambda_{X_5}^{(2)}\right)^* + 4\left(\lambda_{X_9}\right)^*)$	$+2\left(\lambda_{X_5}^{(2)}\right)^*\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 9

Column	Representative term structure
T1	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T2	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$-\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-\lambda_5\,(\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—	$-\lambda_5\,(\lambda_5)^*$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—	$-\lambda_5\,(\lambda_5)^*$
E	-4	3	N/A	3	N/A	2	N/A	$-\frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	—	$-\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 9

Column	Representative term structure
T1	$+8\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(2)}\right)^*$
T2	$+4\lambda_{X_9}\left(\lambda_{X_5}^{(2)}\right)^*$
T3	$-\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(2)}\right)^*$	$+4\lambda_{X_9}\left(\lambda_{X_5}^{(2)}\right)^*$	$-\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	$-\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 9

Column	Representative term structure
T1	$+\frac{51{g_2}^4}{10}$
T2	$-24{g_2}^2\lambda_{12}^{(\times)}$
T3	$-6{g_1}^2\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\frac{51{g_2}^4}{10}$	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-6{g_1}^2\lambda_{12}^{(\times)}$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{51{g_2}^4}{10}$	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-30{g_1}^2\lambda_{12}^{(\times)}$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{51{g_2}^4}{10}$	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-6{g_1}^2\lambda_{12}^{(\times)}$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{51{g_2}^4}{10}$	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-30{g_1}^2\lambda_{12}^{(\times)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 9

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+10{\lambda_{12}^{(\times)}}^{2}$
T3	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+10{\lambda_{12}^{(\times)}}^{2}$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+10{\lambda_{12}^{(\times)}}^{2}$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+10{\lambda_{12}^{(\times)}}^{2}$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+10{\lambda_{12}^{(\times)}}^{2}$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 6 of 9

Column	Representative term structure
T1	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T2	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$
T3	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 7 of 9

Column	Representative term structure
T1	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
T3	$+\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,(y_2)^\dagger\right) + 4\left(\lambda_{X_2}\right)^* + 8\left(\lambda_{X_7}^{(1)}\right)^* + 10\left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$	$+\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,(y_2)^\dagger\right) + 4\left(\lambda_{X_2}\right)^* + 8\left(\lambda_{X_7}^{(1)}\right)^* + 10\left(\lambda_{X_7}^{(2)}\right)^*)$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	—	—

Quantity: $\lambda_{12}^{(\times)}$

Terms 8 of 9

Column	Representative term structure
T1	$+2\left(\lambda_{X_7}^{(2)}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T2	$-\frac{4}{5}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T3	$+4\lambda_{X_2}\left(\lambda_{X_7}^{(2)}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+2\left(\lambda_{X_7}^{(2)}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$-\frac{4}{5}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+4\lambda_{X_2}\left(\lambda_{X_7}^{(2)}\right)^{*}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(\times)}$

Terms 9 of 9

Column	Representative term structure
T1	$+8\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(2)}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(2)}\right)^{*}$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

Quantity: $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
B	−3	2	N/A	2	N/A	1	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
C	1	2	N/A	2	N/A	3	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
C	−1	2	N/A	2	N/A	3	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
D	0	3	N/A	1	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{3}\lambda_5\,(\lambda_5)^*$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{3}\lambda_5\,(\lambda_5)^*$
D	−2	3	N/A	1	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{3}\lambda_5\,(\lambda_5)^*$
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 9

Column	Representative term structure
T1	$+64\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$
T2	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$
T3	$-9{g_2}^2\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+64\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+64\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 9

Column	Representative term structure
T1	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
B	−3	2	N/A	2	N/A	1	N/A	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
C	−3	2	N/A	2	N/A	3	N/A	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
D	2	3	N/A	1	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
D	−2	3	N/A	1	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
E	0	3	N/A	3	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 4 of 9

Column	Representative term structure
T1	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$
T2	$-64\lambda_7\,(\lambda_7)^*$
T3	$+4\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	$-64\lambda_7\,(\lambda_7)^*$	$+4\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	-3	2	N/A	2	N/A	1	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	$-64\lambda_7\,(\lambda_7)^*$	$+4\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	-3	2	N/A	2	N/A	3	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—
E	-4	3	N/A	3	N/A	2	N/A	$+2\lambda^{(A)}_{HS_1}\,\lambda^{(1)}_{S_1}$	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 5 of 9

Column	Representative term structure
T1	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\lambda_{11})^*)$
T2	$+4\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$
T3	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\lambda_{11})^*)$	$+4\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$+{g_1}^2\left(-18{g_2}^2 - 15\lambda_{HS_1}^{(A)}\right)$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\lambda_{11})^*)$	$+4\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	$+{g_1}^2\left(-18{g_2}^2 - 15\lambda_{HS_1}^{(A)}\right)$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 6 of 9

Column	Representative term structure
T1	$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda_{S_1}^{(A)})$
T3	$+\frac{4}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda_{S_1}^{(A)})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda_{S_1}^{(A)})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 7 of 9

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T2	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$
T3	$-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{8}{9}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{4}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 8 of 9

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
T2	$+{g_1}^2\left(4{g_2}^2 - \frac{15\lambda^{(A)}_{HS_1}}{2}\right)$
T3	$+{g_1}^2\left(-4{g_2}^2 - \frac{15\lambda^{(A)}_{HS_1}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$	$+{g_1}^2\left(4{g_2}^2 - \frac{15\lambda^{(A)}_{HS_1}}{2}\right)$	—
D	−2	3	N/A	1	N/A	2	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$	—	$+{g_1}^2\left(-4{g_2}^2 - \frac{15\lambda^{(A)}_{HS_1}}{2}\right)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$	$+{g_1}^2\left(4{g_2}^2 - \frac{15\lambda^{(A)}_{HS_1}}{2}\right)$	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$	—	$+{g_1}^2\left(-4{g_2}^2 - \frac{15\lambda^{(A)}_{HS_1}}{2}\right)$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$	—	$+{g_1}^2\left(-8{g_2}^2 - \frac{51\lambda^{(A)}_{HS_1}}{2}\right)$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 9 of 9

Column	Representative term structure
T1	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

Quantity: $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$
T2	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
A	−4	1	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$
E	2	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$
E	−2	3	N/A	3	N/A	2	N/A	—	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 9

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T2	$+{g_1}^2\left({4g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T3	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$+{g_1}^2\left({4g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{8}{9}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$+{g_1}^2\left({4g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	$+{g_1}^2\left({8g_2}^2-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{4}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 9

Column	Representative term structure
T1	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T2	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
A	-4	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
E	-2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
E	-4	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 4 of 9

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\;(Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\;(Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\;(Y_u)^\dagger\right) + 16\lambda_{S_2}^{(A)})$
T2	$-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\;Y_e\;(Y_e)^\dagger\right)$
T3	$-\frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\;(Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\;(Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\;(Y_u)^\dagger\right) + 16\lambda_{S_2}^{(A)})$	$-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\;Y_e\;(Y_e)^\dagger\right)$	$-\frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\;(Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\;(Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\;(Y_u)^\dagger\right) + 16\lambda_{S_2}^{(A)})$	$-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\;Y_e\;(Y_e)^\dagger\right)$	$-\frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 5 of 9

Column	Representative term structure
T1	$+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T2	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	$+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	−1	2	N/A	2	N/A	1	N/A	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 6 of 9

Column	Representative term structure
T1	$+16\lambda_5 \left(\lambda_5\right)^*$
T2	$+{g_1}^2 \left(18{g_2}^2 - 15\lambda_{HS_2}^{(A)}\right)$
T3	$-9{g_2}^2\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+16\lambda_5 \left(\lambda_5\right)^*$	$+{g_1}^2 \left(18{g_2}^2 - 15\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
B	-1	2	N/A	2	N/A	1	N/A	$+16\lambda_5 \left(\lambda_5\right)^*$	$+{g_1}^2 \left(6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
B	-3	2	N/A	2	N/A	1	N/A	$+16\lambda_5 \left(\lambda_5\right)^*$	—	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+16\lambda_5 \left(\lambda_5\right)^*$	$+{g_1}^2 \left(18{g_2}^2 - 15\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
C	-1	2	N/A	2	N/A	3	N/A	$+16\lambda_5 \left(\lambda_5\right)^*$	$+{g_1}^2 \left(6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
C	-3	2	N/A	2	N/A	3	N/A	$+16\lambda_5 \left(\lambda_5\right)^*$	—	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 7 of 9

Column	Representative term structure
T1	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$
T2	$-4\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	$-4\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$
B	−3	2	N/A	2	N/A	1	N/A	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	$-4\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$
C	−3	2	N/A	2	N/A	3	N/A	$+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 8 of 9

Column	Representative term structure
T1	$+64\lambda_8 \left(\lambda_8\right)^*$
T2	$+\lambda_{X_{17}}^{(1)}(8 \left(\lambda_{X_{17}}^{(1)}\right)^* - 4 \left(\lambda_{10}\right)^*)$
T3	$-64\lambda_{X_{10}} \left(\lambda_{X_{10}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+64\lambda_8 \left(\lambda_8\right)^*$	$+\lambda_{X_{17}}^{(1)}(8 \left(\lambda_{X_{17}}^{(1)}\right)^* - 4 \left(\lambda_{10}\right)^*)$	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	$-64\lambda_{X_{10}} \left(\lambda_{X_{10}}\right)^*$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+64\lambda_8 \left(\lambda_8\right)^*$	$+\lambda_{X_{17}}^{(1)}(8 \left(\lambda_{X_{17}}^{(1)}\right)^* - 4 \left(\lambda_{10}\right)^*)$	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	$-64\lambda_{X_{10}} \left(\lambda_{X_{10}}\right)^*$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 9 of 9

Column	Representative term structure
T1	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$
T2	$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
C	−1	2	N/A	2	N/A	3	N/A	—	$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
C	−3	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$	$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

$\beta^{(1)}_{\lambda_7}$

Quantity: λ_7

$$16\pi^2\,\mu\frac{d\lambda_7}{d\mu} = \beta^{(1)}_{\lambda_7}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-\frac{1}{3}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3	$-9{g_2}^2\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-\frac{1}{3}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$	$-9{g_2}^2\lambda_7$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	$-\frac{1}{2}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$	$-9{g_2}^2\lambda_7$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_7

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda_7$
T2	$+2\lambda_1\lambda_7$
T3	$+8\lambda^{(1)}_{HS_1}\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_7$	$+2\lambda_1\lambda_7$	$+8\lambda^{(1)}_{HS_1}\lambda_7$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_7$	$+2\lambda_1\lambda_7$	$+8\lambda^{(1)}_{HS_1}\lambda_7$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_7

Terms 3 of 4

Column	Representative term structure
T1	$-4\lambda_{HS_1}^{(A)}\lambda_7$
T2	$+2\lambda_7\lambda_{S_1}^{(1)}$
T3	$+4\lambda_8\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-4\lambda_{HS_1}^{(A)}\lambda_7$	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-4\lambda_{HS_1}^{(A)}\lambda_7$	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{12}}$

Quantity: λ_{12}

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu} = \beta^{(1)}_{\lambda_{12}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	$+\lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{12}

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T3	$+\lambda_{12}^{(A)}\left(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+\lambda_{12}^{(A)}\left(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+\lambda_{12}^{(A)}\left(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{12}

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$
T2	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$
T3	$+\lambda^{(A)}_{HS_1}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$	$+\lambda^{(A)}_{HS_1}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$	$+\lambda^{(A)}_{HS_1}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{12}

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$
T2	$+\lambda^{(A)}_{HS_2}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}}\right)$
T3	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}}\right)$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}}\right)$	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda_8}$

Quantity: λ_8

$$16\pi^2\,\mu\frac{d\lambda_8}{d\mu} = \beta^{(1)}_{\lambda_8}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$
T3	$+4\lambda_7\,(\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$	$+4\lambda_7\,(\lambda_9)^*$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	$+4\lambda_7\,(\lambda_9)^*$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_8

Terms 2 of 4

Column	Representative term structure
T1	$-9{g_2}^2\lambda_8$
T2	$-3{g_1}^2\lambda_8$
T3	$+2\lambda_1\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_8$	$-3{g_1}^2\lambda_8$	$+2\lambda_1\lambda_8$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_8$	$-3{g_1}^2\lambda_8$	$+2\lambda_1\lambda_8$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_8

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_8$
T2	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3	$+2\lambda_8\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_8

Terms 4 of 4

Column	Representative term structure
T1	$\lambda_8 (3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_8 (3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(A)}_{S_1}}$

Quantity: $\lambda^{(A)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_1}}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda^{(A)}_{S_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$
T2	$+\lambda_{X_9}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) - \frac{44\left(\lambda_{X_9}\right)^*}{3}\right)$
T3	$-\frac{4}{3}\left(\lambda_{X_9}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda^{(A)}_{S_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$	$+\lambda_{X_9}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) - \frac{44\left(\lambda_{X_9}\right)^*}{3}\right)$	$-\frac{4}{3}\left(\lambda_{X_9}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{S_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$	$+\lambda_{X_9}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_1\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) - \frac{44\left(\lambda_{X_9}\right)^*}{3}\right)$	$-\frac{4}{3}\left(\lambda_{X_9}\right)^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big)$
T2	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_1\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big)$
T3	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big)$	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_1\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big)$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \big(y_1^{\mathrm{conj}}\big)^\dagger\Big)$	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_1\ \big(y_1^{\mathrm{conj}}\big)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big)$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)$
T2	$-\frac{4}{3}\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^{*}$
T3	$-\frac{224}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)$	$-\frac{4}{3}\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^{*}$	$-\frac{224}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{2}{3}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\,y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)$	$-\frac{4}{3}\lambda_{X_5}^{(2)}\left(\lambda_{X_5}^{(1)}\right)^{*}$	$-\frac{224}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$-\frac{8}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^{*}$
T2	$-2{g_2}^4$
T3	$-24{g_2}^2\lambda_{S_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^{*}$	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
D	2	3	N/A	1	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
D	-2	3	N/A	1	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^{*}$	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$+3{\lambda_{HS_1}^{(A)}}^2$
T2	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3	$+20{\lambda_{S_1}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
D	2	3	N/A	1	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
D	−2	3	N/A	1	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	0	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	−2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	−4	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$4\lambda_{S_1}^{(A)} T_\nu^{(1)}$
T2	$+{g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right)$
T3	$-\frac{4}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$4\lambda_{S_1}^{(A)} T_\nu^{(1)}$	$+{g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right)$	—
D	−2	3	N/A	1	N/A	2	N/A	$4\lambda_{S_1}^{(A)} T_\nu^{(1)}$	$+{g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right)$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$-\frac{4}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$
E	2	3	N/A	3	N/A	2	N/A	$4\lambda_{S_1}^{(A)} T_\nu^{(1)}$	$+{g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right)$	—
E	−2	3	N/A	3	N/A	2	N/A	$4\lambda_{S_1}^{(A)} T_\nu^{(1)}$	$+{g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right)$	—
E	−4	3	N/A	3	N/A	2	N/A	$4\lambda_{S_1}^{(A)} T_\nu^{(1)}$	$+{g_1}^2 \left(32{g_2}^2 - 48\lambda_{S_1}^{(A)}\right)$	—

Quantity: $\lambda_{S_1}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$+12{\lambda_{12}^{(A)}}^2$
T2	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$
T3	$-\frac{4}{3}\lambda_{X_7}^{(2)}\left(\lambda_{X_7}^{(2)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	—
E	2	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	—
E	-2	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	$-\frac{4}{3}\lambda_{X_7}^{(2)}\left(\lambda_{X_7}^{(2)}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	—

$\beta^{(1)}_{\lambda_{11}}$

Quantity: λ_{11}

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu} = \beta^{(1)}_{\lambda_{11}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{11}(3T_\nu^{(1)} + T_\nu^{(2)})$
T2	$+4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$
T3	$-9{g_2}^2\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda_{11}(3T_\nu^{(1)} + T_\nu^{(2)})$	$+4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$	$-9{g_2}^2\lambda_{11}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$	$-9{g_2}^2\lambda_{11}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{11}

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{11}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_{11}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_{11}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{11}

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$
T2	$+\lambda^{(A)}_{HS_1}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} + \frac{\lambda^{(1)}_{X_{17}}}{2}\right)$
T3	$+12\lambda^{(1)}_{S_1}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(A)}_{HS_1}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} + \frac{\lambda^{(1)}_{X_{17}}}{2}\right)$	$+12\lambda^{(1)}_{S_1}\lambda_{11}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(A)}_{HS_1}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} + \frac{\lambda^{(1)}_{X_{17}}}{2}\right)$	$+12\lambda^{(1)}_{S_1}\lambda_{11}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{11}

Terms 4 of 4

Column	Representative term structure
T1	$\lambda_{11}\left(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_{11}\left(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_9}$

Quantity: λ_9

$$16\pi^2\,\mu\frac{d\lambda_9}{d\mu} = \beta^{(1)}_{\lambda_9}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_9(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$+\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$
T3	$+4\lambda_7\,(\lambda_8)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$\lambda_9(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$	$+4\lambda_7\,(\lambda_8)^*$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	$+\frac{5}{6}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$	$+4\lambda_7\,(\lambda_8)^*$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 2 of 5

Column	Representative term structure
T1	$-9{g_2}^2\lambda_9$
T2	$-3{g_1}^2\lambda_9$
T3	$+8\lambda_{12}^{(1)}\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_9$	$-3{g_1}^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_9$	$-3{g_1}^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 3 of 5

Column	Representative term structure
T1	$-4\lambda_{12}^{(A)}\lambda_9$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$
T3	$+2\lambda_{S_1}^{(1)}\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-4\lambda_{12}^{(A)}\lambda_9$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$	$+2\lambda_{S_1}^{(1)}\lambda_9$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-4\lambda_{12}^{(A)}\lambda_9$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$	$+2\lambda_{S_1}^{(1)}\lambda_9$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_9\lambda_{S_2}^{(1)}$
T2	$+5{\lambda_{10}}^2$
T3	$+2\lambda_{10}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5{\lambda_{10}}^2$ $+5{\lambda_{11}}^2$	$+2\lambda_{10}\lambda_{11}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5{\lambda_{10}}^2$ $+5{\lambda_{11}}^2$	$+2\lambda_{10}\lambda_{11}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 5 of 5

Column	Representative term structure
T1	$+{\lambda_{X_{17}}^{(1)}}^2$
T2	$\lambda_9(3T_\nu^{(1)} + 3T_\nu^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+{\lambda_{X_{17}}^{(1)}}^2$	—
B	-3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+{\lambda_{X_{17}}^{(1)}}^2$	$\lambda_9(3T_\nu^{(1)} + 3T_\nu^{(2)})$
C	-3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{10}}$

Quantity: λ_{10}

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu} = \beta^{(1)}_{\lambda_{10}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$
T2	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$
T3	$-9{g_2}^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-9{g_2}^2\lambda_{10}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-9{g_2}^2\lambda_{10}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{10}

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{10}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{10}

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$
T2	$+\lambda^{(A)}_{HS_2}\left(-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} - \frac{\lambda^{(1)}_{X_{17}}}{2}\right)$
T3	$+12\lambda_{10}\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{HS_2}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(A)}_{HS_2}\left(-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} - \frac{\lambda^{(1)}_{X_{17}}}{2}\right)$	$+12\lambda_{10}\lambda^{(1)}_{S_2}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{HS_2}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(A)}_{HS_2}\left(-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} - \frac{\lambda^{(1)}_{X_{17}}}{2}\right)$	$+12\lambda_{10}\lambda^{(1)}_{S_2}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{10}

Terms 4 of 4

Column	Representative term structure
T1	$\lambda_{10}\left(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_{10}\left(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(A)}_{S_2}}$

Quantity: $\lambda^{(A)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_2}}$$

Terms 1 of 7

Column	Representative term structure
T1	$4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$
T2	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$-2{g_2}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$
A	−2	1	N/A	3	N/A	2	N/A	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$
A	−4	1	N/A	3	N/A	2	N/A	$4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$
E	2	3	N/A	3	N/A	2	N/A	$4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$
E	−4	3	N/A	3	N/A	2	N/A	$4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-2{g_2}^4$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(A)}\right)$
T2	$-24{g_2}^2\lambda_{S_2}^{(A)}$
T3	$+3{\lambda_{HS_2}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(A)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$
A	-2	1	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$
A	-4	1	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(A)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(A)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(32{g_2}^2-48\lambda_{S_2}^{(A)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_2}^{(A)}\right)$	$-24{g_2}^2\lambda_{S_2}^{(A)}$	$+3{\lambda_{HS_2}^{(A)}}^2$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T2	$+20{\lambda_{S_2}^{(A)}}^2$
T3	$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	—
A	−2	1	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
A	−4	1	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	—
E	−2	3	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
E	−4	3	N/A	3	N/A	2	N/A	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$-\frac{4}{3}\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
T2	$+\lambda_{X_2}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)-\frac{44\left(\lambda_{X_2}\right)^{*}}{3}\right)$
T3	$+\frac{76}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{4}{3}\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\lambda_{X_2}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)-\frac{44\left(\lambda_{X_2}\right)^{*}}{3}\right)$	$+\frac{76}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{4}{3}\left(\lambda_{X_2}\right)^{*}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$	$+\lambda_{X_2}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2\right)^{\dagger}\right)-\frac{44\left(\lambda_{X_2}\right)^{*}}{3}\right)$	$+\frac{76}{45}\,\mathrm{Tr}\left(y_2\,\left(y_2\right)^{\dagger}\,y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_2\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\ y_2^{\mathrm{conj}}\ (y_2)^{\dagger}\Big)$
T2	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\ y_2^{\mathrm{conj}}\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\Big)$
T3	$-\frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_2\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\ y_2^{\mathrm{conj}}\ (y_2)^{\dagger}\Big)$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\ y_2^{\mathrm{conj}}\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\Big)$	$-\frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{76}{45}\,\mathrm{Tr}\Big(y_2\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\ y_2^{\mathrm{conj}}\ (y_2)^{\dagger}\Big)$	$-\frac{2}{3}\,\mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\ y_2^{\mathrm{conj}}\ \big(y_2^{\mathrm{conj}}\big)^{\dagger}\Big)$	$-\frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^{*}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$-\frac{4}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
T2	$-\frac{8}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$
T3	$-\frac{4}{3}\lambda_{X_5}^{(2)}\left(\lambda_{X_5}^{(2)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{4}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$	$-\frac{8}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$-\frac{4}{3}\lambda_{X_5}^{(2)}\left(\lambda_{X_5}^{(2)}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{4}{3}\lambda_{X_7}^{(2)}\left(\lambda_{X_7}^{(1)}\right)^*$	$-\frac{8}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$+12{\lambda_{12}^{(A)}}^2$
T2	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$
T3	$-\frac{4}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	—
E	2	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	—
E	-2	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	$-\frac{4}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
E	-4	3	N/A	3	N/A	2	N/A	$+12{\lambda_{12}^{(A)}}^2$	$-\frac{4{\lambda_{12}^{(\times)}}^2}{3}$	—

$\beta^{(1)}_{\lambda_{X_1}}$

Quantity: λ_{X_1}

$$16\pi^2\,\mu\frac{d\lambda_{X_1}}{d\mu} = \beta^{(1)}_{\lambda_{X_1}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{X_1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
T2	$+2\lambda_{X_2}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$-\frac{23}{15}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\,y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda_{X_1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$	$+2\lambda_{X_2}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-\frac{23}{15}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\,y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{X_2}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda_{X_1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$	$+2\lambda_{X_2}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-\frac{23}{15}\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\,y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_1}

Terms 2 of 5

Column	Representative term structure
T1	$-24{g_2}^2\lambda_{X_1}$
T2	$+12\lambda_{S_2}^{(1)}\lambda_{X_1}$
T3	$-8\lambda_{S_2}^{(A)}\lambda_{X_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{X_1}$	$+12\lambda_{S_2}^{(1)}\lambda_{X_1}$	$-8\lambda_{S_2}^{(A)}\lambda_{X_1}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+12\lambda_{S_2}^{(1)}\lambda_{X_1}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{X_1}$	$+12\lambda_{S_2}^{(1)}\lambda_{X_1}$	$-8\lambda_{S_2}^{(A)}\lambda_{X_1}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_1}

Terms 3 of 5

Column	Representative term structure
T1	$+11{\lambda_{X_2}}^2$
T2	$+{\lambda_{X_7}^{(1)}}^2$
T3	$+2{\lambda_{X_{16}}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+11{\lambda_{X_2}}^2$	$+{\lambda_{X_7}^{(1)}}^2$	$+2{\lambda_{X_{16}}}^2$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+9{\lambda_{X_2}}^2$	—	$+2{\lambda_{X_{16}}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+11{\lambda_{X_2}}^2$	$+{\lambda_{X_7}^{(2)}}^2$	$+2{\lambda_{X_{16}}}^2$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_1}

Terms 4 of 5

Column	Representative term structure
T1	$\lambda_{X_1}(8T_\nu^{(2)} + 8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
T2	$-4\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$+3{\lambda_{X_7}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$\lambda_{X_1}(8T_\nu^{(2)} + 8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$	$-4\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$+3{\lambda_{X_7}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{X_7}^{(1)}}^2$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_1}

Terms 5 of 5

Column	Representative term structure
T1	$+2\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_2}}$

Quantity: λ_{X_2}

$$16\pi^2\,\mu\frac{d\lambda_{X_2}}{d\mu} = \beta^{(1)}_{\lambda_{X_2}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{S_2}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 22\lambda_{X_2})$
T2	$+\lambda_{X_1}(8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 88\left(\lambda_{X_2}\right)^*)$
T3	$+\lambda_{X_2}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 22\lambda_{X_2})$	$+\lambda_{X_1}(8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 88\left(\lambda_{X_2}\right)^*)$	$+\lambda_{X_2}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$\lambda^{(1)}_{S_2}(4\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 18\lambda_{X_2})$	$+\lambda_{X_1}(16\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 72\left(\lambda_{X_2}\right)^*)$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{S_2}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 22\lambda_{X_2})$	$+\lambda_{X_1}(8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 88\left(\lambda_{X_2}\right)^*)$	$+\lambda_{X_2}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_2}

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{46}{15}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T2	$-\frac{46}{15}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$-24{g_2}^2\lambda_{X_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{46}{15}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-\frac{46}{15}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-24{g_2}^2\lambda_{X_2}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{46}{15}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-\frac{46}{15}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-24{g_2}^2\lambda_{X_2}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_2}

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$
T2	$+4\lambda_{HS_2}^{(1)}\lambda_{X16}$
T3	$+\lambda_{X_2}(8T_\nu^{(2)} + 8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X16}$	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X16}$	$+\lambda_{X_2}(8T_\nu^{(2)} + 8\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_2}^{(1)}\lambda_{X16}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_2}

Terms 4 of 5

Column	Representative term structure
T1	$-8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T2	$-8\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
T3	$+\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	$-8\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	$+\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_2}

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)} + 4\lambda_{X_7}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)} + 4\lambda_{X_7}^{(2)})$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{X_5}}$

Quantity: $\lambda^{(1)}_{X_5}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{X_5}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{X_5}}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda^{(1)}_{12}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 8\lambda^{(1)}_{X_5} + 6\lambda_{X_9})$
T2	$+\lambda^{(1)}_{X_5}(4T^{(1)}_\nu + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T^{(2)}_\nu)$
T3	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{12}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 8\lambda^{(1)}_{X_5} + 6\lambda_{X_9})$	$+\lambda^{(1)}_{X_5}(4T^{(1)}_\nu + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T^{(2)}_\nu)$	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda^{(1)}_{12}(2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 8\lambda^{(1)}_{X_5} + 10\lambda_{X_9})$	—	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{12}(2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 8\lambda^{(1)}_{X_5} + 10\lambda_{X_9})$	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 2 of 7

Column	Representative term structure
T1	$+4\left(\lambda_5\right)^*\left(\lambda_{X_{14}}\right)^*$
T2	$+24\lambda_{X_6}\left(\lambda^{(1)}_{X_5}\right)^*$
T3	$-12{g_2}^2\lambda^{(1)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\left(\lambda_5\right)^*\left(\lambda_{X_{14}}\right)^*$	$+24\lambda_{X_6}\left(\lambda^{(1)}_{X_5}\right)^*$	$-12{g_2}^2\lambda^{(1)}_{X_5}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\left(\lambda_5\right)^*\left(\lambda_{X_{14}}\right)^*$	$+40\lambda_{X_6}\left(\lambda^{(1)}_{X_5}\right)^*$	$-12{g_2}^2\lambda^{(1)}_{X_5}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\left(\lambda_5\right)^*\left(\lambda_{X_{14}}\right)^*$	—	$-24{g_2}^2\lambda^{(1)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_5}^{(1)}$

Terms 3 of 7

Column	Representative term structure
T1	$-6{g_1}^2\lambda_{X_5}^{(1)}$
T2	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$
T3	$+2\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{X_5}^{(1)}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	$+2\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-6{g_1}^2\lambda_{X_5}^{(1)}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	$+2\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{X_5}^{(1)}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	$+2\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 4 of 7

Column	Representative term structure
T1	$+8\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$
T2	$+4\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$
T3	$+\lambda^{(1)}_{X_5}(2T^{(1)}_{\nu}+2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)+4T^{(2)}_{\nu})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	$+4\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	—	$+\lambda^{(1)}_{X_5}(2T^{(1)}_{\nu}+2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)+4T^{(2)}_{\nu})$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_5}^{(1)}$

Terms 5 of 7

Column	Representative term structure
T1	$-4\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$
T2	$+\lambda_{X_5}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)})$
T3	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-4\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+\lambda_{X_5}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)})$	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 6 of 7

Column	Representative term structure
T1	$+8\lambda_{X_6}(5\left(\lambda^{(1)}_{X_5}\right)^* + \left(\lambda^{(2)}_{X_5}\right)^*)$
T2	$+\lambda^{(\times)}_{12}(4\lambda^{(2)}_{X_5} + 4\lambda_{X_9})$
T3	$+\lambda^{(A)}_{S_1}(-4\lambda^{(1)}_{X_5} - 2\lambda^{(2)}_{X_5})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_6}(5\left(\lambda^{(1)}_{X_5}\right)^* + \left(\lambda^{(2)}_{X_5}\right)^*)$	$+\lambda^{(\times)}_{12}(4\lambda^{(2)}_{X_5} + 4\lambda_{X_9})$	$+\lambda^{(A)}_{S_1}(-4\lambda^{(1)}_{X_5} - 2\lambda^{(2)}_{X_5})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 7 of 7

Column	Representative term structure
T1	$+\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$
T2	$+\lambda^{(A)}_{S_2}(4\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	$+\lambda^{(A)}_{S_2}(4\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(2)}_{X_5}}$

Quantity: $\lambda^{(2)}_{X_5}$

$$16\pi^2\,\mu\frac{d\lambda^{(2)}_{X_5}}{d\mu} = \beta^{(1)}_{\lambda^{(2)}_{X_5}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\;(y_1)^\dagger\right) + 16\lambda^{(1)}_{X_5} + 20\lambda^{(2)}_{X_5} + 8\lambda_{X_9})$
T2	$+\lambda^{(2)}_{X_5}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\;\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)})$
T3	$-\frac{8}{5}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\;(y_1)^\dagger\;y_2\;(y_2)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\;(y_1)^\dagger\right) + 16\lambda^{(1)}_{X_5} + 20\lambda^{(2)}_{X_5} + 8\lambda_{X_9})$	$+\lambda^{(2)}_{X_5}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\;\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)})$	$-\frac{8}{5}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\;(y_1)^\dagger\;y_2\;(y_2)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_5}^{(2)}$

Terms 2 of 4

Column	Representative term structure
T1	$-2\left(\lambda_5\right)^*\left(\lambda_{X_{14}}\right)^*$
T2	$+16\lambda_{X_6}\left(\lambda_{X_5}^{(2)}\right)^*$
T3	$-24{g_2}^2\lambda_{X_5}^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-2\left(\lambda_5\right)^*\left(\lambda_{X_{14}}\right)^*$	$+16\lambda_{X_6}\left(\lambda_{X_5}^{(2)}\right)^*$	$-24{g_2}^2\lambda_{X_5}^{(2)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_5}$

Terms 3 of 4

Column	Representative term structure
T1	$-6{g_1}^2\lambda^{(2)}_{X_5}$
T2	$+8\lambda^{(1)}_{12}\lambda^{(2)}_{X_5}$
T3	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda^{(2)}_{X_5}$	$+8\lambda^{(1)}_{12}\lambda^{(2)}_{X_5}$	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_5}$

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$
T2	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$
T3	$-2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$	$-2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda_{X_6}}$

Quantity: λ_{X_6}

$$16\pi^2\,\mu\frac{d\lambda_{X_6}}{d\mu} = \beta^{(1)}_{\lambda_{X_6}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right))$
T2	$+4\lambda_{X_9}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right))$	$+4\lambda_{X_9}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+2\lambda_{X_9}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+2\lambda_{X_9}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_6}

Terms 2 of 5

Column	Representative term structure
T1	$+12\lambda_{S_1}^{(1)}\lambda_{X_6}$
T2	$+3{\lambda_{X_5}^{(1)}}^2$
T3	$+9{\lambda_{X_9}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+12\lambda_{S_1}^{(1)}\lambda_{X_6}$	$+3{\lambda_{X_5}^{(1)}}^2$	$+9{\lambda_{X_9}}^2$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+12\lambda_{S_1}^{(1)}\lambda_{X_6}$	—	$+11{\lambda_{X_9}}^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda_{S_1}^{(1)}\lambda_{X_6}$	$+3{\lambda_{X_5}^{(1)}}^2$	$+11{\lambda_{X_9}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_6}

Terms 3 of 5

Column	Representative term structure
T1	$+2{\lambda_{X_{18}}}^2$
T2	$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right))$
T3	$-\frac{23}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2{\lambda_{X_{18}}}^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2{\lambda_{X_{18}}}^2$	$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right))$	$-\frac{23}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2{\lambda_{X_{18}}}^2$	$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right))$	$-\frac{23}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\,y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_6}

Terms 4 of 5

Column	Representative term structure
T1	$-24{g_2}^2\lambda_{X_6}$
T2	$-8\lambda_{S_1}^{(A)}\lambda_{X_6}$
T3	$+{\lambda_{X_5}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-24{g_2}^2\lambda_{X_6}$	$-8\lambda_{S_1}^{(A)}\lambda_{X_6}$	$+{\lambda_{X_5}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{X_6}$	$-8\lambda_{S_1}^{(A)}\lambda_{X_6}$	$+{\lambda_{X_5}^{(2)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_6}

Terms 5 of 5

Column	Representative term structure
T1	$+2\lambda^{(1)}_{X_5}\lambda^{(2)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{X_5}\lambda^{(2)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{X_7}}$

Quantity: $\lambda^{(1)}_{X_7}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{X_7}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{X_7}}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda^{(1)}_{12}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda^{(1)}_{X_7})$
T2	$+\lambda^{(1)}_{X_7}(4T^{(1)}_\nu + 2T^{(2)}_\nu + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
T3	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{12}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda^{(1)}_{X_7})$	$+\lambda^{(1)}_{X_7}(4T^{(1)}_\nu + 2T^{(2)}_\nu + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right))$	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$\lambda^{(1)}_{12}(4\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\lambda_{X_2} + 8\lambda^{(1)}_{X_7})$	—	$-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{12}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda^{(1)}_{X_7})$	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_7}^{(1)}$

Terms 2 of 7

Column	Representative term structure
T1	$+40\lambda_{X_1}\left(\lambda_{X_7}^{(1)}\right)^*$
T2	$-12{g_2}^2\lambda_{X_7}^{(1)}$
T3	$-6{g_1}^2\lambda_{X_7}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+40\lambda_{X_1}\left(\lambda_{X_7}^{(1)}\right)^*$	$-12{g_2}^2\lambda_{X_7}^{(1)}$	$-6{g_1}^2\lambda_{X_7}^{(1)}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+24\lambda_{X_1}\left(\lambda_{X_7}^{(1)}\right)^*$	$-12{g_2}^2\lambda_{X_7}^{(1)}$	$-6{g_1}^2\lambda_{X_7}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{X_7}^{(1)}$	$-6{g_1}^2\lambda_{X_7}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_7}$

Terms 3 of 7

Column	Representative term structure
T1	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$
T2	$+4\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$
T3	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	$+4\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	$+8\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	—	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_7}^{(1)}$

Terms 4 of 7

Column	Representative term structure
T1	$-4\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$
T2	$+4\lambda_5\lambda_{X_{12}}$
T3	$+\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	$+4\lambda_5\lambda_{X_{12}}$	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_5\lambda_{X_{12}}$	$+\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	$+2\lambda_5\lambda_{X_{12}}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_7}^{(1)}$

Terms 5 of 7

Column	Representative term structure
T1	$+4\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$
T2	$+\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
T3	$-\frac{22}{15}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$+\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right))\ -\frac{22}{15}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$	
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_7}$

Terms 6 of 7

Column	Representative term structure
T1	$+8\lambda_{X_1}(5\left(\lambda^{(1)}_{X_7}\right)^* + \left(\lambda^{(2)}_{X_7}\right)^*)$
T2	$+\lambda^{(\times)}_{12}(4\lambda_{X_2} + 4\lambda^{(2)}_{X_7})$
T3	$+\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_1}(5\left(\lambda^{(1)}_{X_7}\right)^* + \left(\lambda^{(2)}_{X_7}\right)^*)$	$+\lambda^{(\times)}_{12}(4\lambda_{X_2} + 4\lambda^{(2)}_{X_7})$	$+\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_7}$

Terms 7 of 7

Column	Representative term structure
T1	$+\lambda^{(A)}_{S_1}(4\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$
T2	$+\lambda^{(A)}_{S_2}(-4\lambda^{(1)}_{X_7} - 2\lambda^{(2)}_{X_7})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda^{(A)}_{S_1}(4\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$	$+\lambda^{(A)}_{S_2}(-4\lambda^{(1)}_{X_7} - 2\lambda^{(2)}_{X_7})$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(2)}_{X_7}}$

Quantity: $\lambda^{(2)}_{X_7}$

$$16\pi^2\,\mu\frac{d\lambda^{(2)}_{X_7}}{d\mu} = \beta^{(1)}_{\lambda^{(2)}_{X_7}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{X_2} + 16\lambda^{(1)}_{X_7} + 20\lambda^{(2)}_{X_7})$
T2	$+\lambda^{(2)}_{X_7}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
T3	$-\frac{8}{5}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}\left(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{X_2} + 16\lambda^{(1)}_{X_7} + 20\lambda^{(2)}_{X_7})$	$+\lambda^{(2)}_{X_7}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$	$-\frac{8}{5}\,\mathrm{Tr}\left(y_1\,\left(y_1\right)^\dagger\,y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_7}^{(2)}$

Terms 2 of 4

Column	Representative term structure
T1	$+16\lambda_{X_1}\left(\lambda_{X_7}^{(2)}\right)^*$
T2	$-24{g_2}^2\lambda_{X_7}^{(2)}$
T3	$-6{g_1}^2\lambda_{X_7}^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+16\lambda_{X_1}\left(\lambda_{X_7}^{(2)}\right)^*$	$-24{g_2}^2\lambda_{X_7}^{(2)}$	$-6{g_1}^2\lambda_{X_7}^{(2)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_7}$

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$
T2	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_7}$
T3	$-2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_7}$	$-2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_7}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_7}$

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_7}$
T2	$+2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_7}$
T3	$-2\lambda_5\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_7}$	$+2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_7}$	$-2\lambda_5\lambda_{X_{12}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda_{X_9}}$

Quantity: λ_{X_9}

$$16\pi^2\,\mu\frac{d\lambda_{X_9}}{d\mu} = \beta^{(1)}_{\lambda_{X_9}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 18\lambda_{X_9})$
T2	$+\lambda_{X_6}(16\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 72\left(\lambda_{X_9}\right)^*)$
T3	$+\lambda_{X_9}(8T_\nu^{(1)} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 18\lambda_{X_9})$	$+\lambda_{X_6}(16\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 72\left(\lambda_{X_9}\right)^*)$	$+\lambda_{X_9}(8T_\nu^{(1)} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right))$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 22\lambda_{X_9})$	$+\lambda_{X_6}(8\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 88\left(\lambda_{X_9}\right)^*)$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 22\lambda_{X_9})$	$+\lambda_{X_6}(8\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 88\left(\lambda_{X_9}\right)^*)$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_9}

Terms 2 of 5

Column	Representative term structure
T1	$-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$
T2	$-8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$
T3	$+6\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\right)$	$-8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	$+6\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$+2\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_9}

Terms 3 of 5

Column	Representative term structure
T1	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{18}}$
T2	$+\lambda_{X_9}(4T^{(1)}_{\nu}+4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$
T3	$-\frac{46}{15}\,\mathrm{Tr}\left(y_1\,(y_1)^{\dagger}\,y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{18}}$	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{18}}$	$+\lambda_{X_9}(4T^{(1)}_{\nu}+4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$	$-\frac{46}{15}\,\mathrm{Tr}\left(y_1\,(y_1)^{\dagger}\,y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{18}}$	$+\lambda_{X_9}(4T^{(1)}_{\nu}+4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))$	$-\frac{46}{15}\,\mathrm{Tr}\left(y_1\,(y_1)^{\dagger}\,y_1^{\mathrm{conj}}\,(y_1)^{\dagger}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_9}

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{46}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$
T2	$-24{g_2}^2\lambda_{X_9}$
T3	$+\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{46}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	$-24{g_2}^2\lambda_{X_9}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{46}{15}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)$	$-24{g_2}^2\lambda_{X_9}$	$+\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_9}

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_{10}}}$

Quantity: $\lambda_{X_{10}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{10}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{10}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-9{g_2}^2\lambda_{X_{10}}$
T3	$-3{g_1}^2\lambda_{X_{10}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_{X_{10}}$	$-3{g_1}^2\lambda_{X_{10}}$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{X_{10}}$	$-3{g_1}^2\lambda_{X_{10}}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{10}}$

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda_{X_{10}}$
T2	$+8\lambda^{(1)}_{HS_2}\lambda_{X_{10}}$
T3	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{10}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{X_{10}}$	$+8\lambda^{(1)}_{HS_2}\lambda_{X_{10}}$	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{10}}$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{X_{10}}$	$+8\lambda^{(1)}_{HS_2}\lambda_{X_{10}}$	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{10}}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{10}}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{S_2}^{(1)}\lambda_{X_{10}}$
T2	$\lambda_{X_{10}}(3T_{\nu}^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	$+2\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	$+2\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	$\lambda_{X_{10}}(3T_{\nu}^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{X_{12}}}$

Quantity: $\lambda_{X_{12}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{12}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{12}}}$$

Terms 1 of 6

Column	Representative term structure
T1	$2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\Big)$
T2	$+\lambda_{X_{12}}(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\Big(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$
T3	$+4\lambda_{X_7}^{(1)}\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\Big)$	$+\lambda_{X_{12}}(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\Big(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+4\lambda_{X_7}^{(1)}\,(\lambda_5)^*$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$4\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\Big)$	—	$+4\lambda_{X_7}^{(1)}\,(\lambda_5)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\,\left(y_2^{\mathrm{conj}}\right)^\dagger\Big)$	—	$+4\lambda_{X_7}^{(1)}\,(\lambda_5)^*$ $-2\lambda_{X_7}^{(2)}\,(\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 2 of 6

Column	Representative term structure
T1	$+8\lambda_{X_{16}}\left(\lambda_5\right)^*$
T2	$-\frac{21{g_2}^2\lambda_{X_{12}}}{2}$
T3	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_5\right)^*$	$-\frac{21{g_2}^2\lambda_{X_{12}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_5\right)^*$	$-\frac{21{g_2}^2\lambda_{X_{12}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_5\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{12}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 3 of 6

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$
T2	$+2\lambda_{1}\lambda_{X_{12}}$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$	$+2\lambda_{1}\lambda_{X_{12}}$	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$	$+2\lambda_{1}\lambda_{X_{12}}$	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$	$+2\lambda_{1}\lambda_{X_{12}}$	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 4 of 6

Column	Representative term structure
T1	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$
T2	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{12}}$
T3	$+\lambda_{X_{12}}(T^{(1)}_{\nu}+2T^{(2)}_{\nu}+2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^{\dagger}\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^{\dagger}\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^{\dagger}\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{12}}$	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$	—	$+\lambda_{X_{12}}(T^{(1)}_{\nu}+2T^{(2)}_{\nu}+2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^{\dagger}\right)+6\,\mathrm{Tr}\left(Y_d\,(Y_d)^{\dagger}\right)+2\,\mathrm{Tr}\left(Y_e\,(Y_e)^{\dagger}\right)+6\,\mathrm{Tr}\left(Y_u\,(Y_u)^{\dagger}\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$	$-2\lambda^{(A)}_{HS_2}\lambda_{X_{12}}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 5 of 6

Column	Representative term structure
T1	$-4\lambda_{HS_1}^{(A)}\,\lambda_{X_{12}}$
T2	$+\lambda_{X_{12}}(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right))$
T3	$-2\lambda_{12}^{(A)}\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-4\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-2\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	$+\lambda_{X_{12}}(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right))$	$-2\lambda_{12}^{(A)}\lambda_{X_{12}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X12}

Terms 6 of 6

Column	Representative term structure
T1	$-2\lambda_{12}^{(\times)}\lambda_{X12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{X12}$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_{14}}}$

Quantity: $\lambda_{X_{14}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{14}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{14}}}$$

Terms 1 of 6

Column	Representative term structure
T1	$4\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_1\,\Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big)$
T2	$+\lambda_{X_{14}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\,\Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$
T3	$+4\,(\lambda_5)^*\,(\Big(\lambda^{(1)}_{X_5}\Big)^* + 2\,\Big(\lambda_{X_{18}}\Big)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$4\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_1\,\Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big)$	$+\lambda_{X_{14}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\,\Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+4\,(\lambda_5)^*\,(\Big(\lambda^{(1)}_{X_5}\Big)^* + 2\,\Big(\lambda_{X_{18}}\Big)^*)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_1\,\Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big)$	—	$+4\,(\lambda_5)^*\,(\Big(\lambda^{(1)}_{X_5}\Big)^* + 2\,\Big(\lambda_{X_{18}}\Big)^*)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_1\,\Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big)$	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 2 of 6

Column	Representative term structure
T1	$-\frac{21{g_2}^2\lambda_{X_{14}}}{2}$
T2	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$
T3	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_{X_{14}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_{X_{14}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{X_{14}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 3 of 6

Column	Representative term structure
T1	$+2\lambda_1\lambda_{X_{14}}$
T2	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{14}}$
T3	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{X_{14}}$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{14}}$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{14}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_{X_{14}}$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{14}}$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{14}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{X_{14}}$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{14}}$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{14}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 4 of 6

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$
T2	$+\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,(Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,(Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,(Y_u)^\dagger\right))$
T3	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}}\left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\,(Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,(Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,(Y_u)^\dagger\right))$	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 5 of 6

Column	Representative term structure
T1	$+\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+2\,(\lambda_5)^*\,(2\left(\lambda_{X_5}^{(1)}\right)^* - \left(\lambda_{X_5}^{(2)}\right)^* + 4\left(\lambda_{X_{18}}\right)^*)$
T3	$-2\lambda_{12}^{(A)}\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+2\,(\lambda_5)^*\,(2\left(\lambda_{X_5}^{(1)}\right)^* - \left(\lambda_{X_5}^{(2)}\right)^* + 4\left(\lambda_{X_{18}}\right)^*)$	$-2\lambda_{12}^{(A)}\lambda_{X_{14}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 6 of 6

Column	Representative term structure
T1	$-2\lambda_{12}^{(\times)}\lambda_{X_{14}}$
T2	$-\frac{4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}}{3}$
T3	$-\frac{4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{X_{14}}$	$-\frac{4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}}{3}$	$-\frac{4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}}{3}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda_{X_{15}}}$

Quantity: $\lambda_{X_{15}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{15}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{15}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{X_{15}}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-9{g_2}^2\lambda_{X_{15}}$
T3	$-3{g_1}^2\lambda_{X_{15}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda_{X_{15}}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_{X_{15}}$	$-3{g_1}^2\lambda_{X_{15}}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{X_{15}}$	$-3{g_1}^2\lambda_{X_{15}}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{15}}$

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda_{X_{15}}$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$
T3	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{X_{15}}$	$+8\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{X_{15}}$	$+8\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{15}}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{S_1}^{(1)}\lambda_{X_{15}}$
T2	$\lambda_{X_{15}}(3T_{\nu}^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	$\lambda_{X_{15}}(3T_{\nu}^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{X_{16}}}$

Quantity: $\lambda_{X_{16}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{16}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{16}}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda_{X_{16}})$
T2	$+\lambda_{X_{16}}(2T^{(2)}_\nu + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$-2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda_{X_{16}})$	$+\lambda_{X_{16}}(2T^{(2)}_\nu + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\lambda_{X_2} + 8\lambda_{X_{16}})$	—	$-4\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda_{X_{16}})$	$+\lambda_{X_{16}}(2T^{(2)}_\nu + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X16}

Terms 2 of 5

Column	Representative term structure
T1	$+40\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$
T2	$-\frac{33{g_2}^2\lambda_{X_{16}}}{2}$
T3	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+40\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{16}}}{2}$	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+24\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$	$-\frac{9{g_2}^2\lambda_{X_{16}}}{2}$	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+40\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{16}}}{2}$	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{16}}$

Terms 3 of 5

Column	Representative term structure
T1	$+6\lambda_1\lambda_{X_{16}}$
T2	$+2\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$
T3	$+2\lambda^{(1)}_{S_2}\lambda_{X_{16}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+6\lambda_1\lambda_{X_{16}}$	$+2\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	$+2\lambda^{(1)}_{S_2}\lambda_{X_{16}}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+6\lambda_1\lambda_{X_{16}}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	$+2\lambda^{(1)}_{S_2}\lambda_{X_{16}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+6\lambda_1\lambda_{X_{16}}$	—	$+2\lambda^{(1)}_{S_2}\lambda_{X_{16}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{16}}$

Terms 4 of 5

Column	Representative term structure
T1	$-4\lambda_{S_2}^{(A)}\lambda_{X_{16}}$
T2	$+4\lambda_5\lambda_{X_{12}}$
T3	$+\lambda_{X_{16}}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\lambda_{X_{16}}$	$+4\lambda_5\lambda_{X_{12}}$	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+12\lambda_5\lambda_{X_{12}}$	$+\lambda_{X_{16}}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\,\left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\lambda_{X_{16}}$	$+4\lambda_5\lambda_{X_{12}}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{16}}$

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{X_{17}}}$

Quantity: $\lambda^{(1)}_{X_{17}}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{X_{17}}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{X_{17}}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{X_{17}}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\lambda_9(8\left(\lambda^{(1)}_{X_{17}}\right)^* - 4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$
T3	$-9{g_2}^2\lambda^{(1)}_{X_{17}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{X_{17}}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\lambda_9(8\left(\lambda^{(1)}_{X_{17}}\right)^* - 4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$	$-9{g_2}^2\lambda^{(1)}_{X_{17}}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	$+\lambda_9(8\left(\lambda^{(1)}_{X_{17}}\right)^* - 4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$	$-9{g_2}^2\lambda^{(1)}_{X_{17}}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{17}}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{X_{17}}^{(1)}$
T2	$+2\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$
T3	$+\lambda_{12}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_{X_{17}}^{(1)}$	$+2\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	$+\lambda_{12}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_{X_{17}}^{(1)}$	$+2\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	$+\lambda_{12}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{17}}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$
T2	$+\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$
T3	$+\lambda_{HS_1}^{(A)}\left(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	$+\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	$+\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_{17}}$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$
T2	$+\lambda^{(A)}_{HS_2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}}\right)$
T3	$\lambda^{(1)}_{X_{17}}\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}}\right)$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}}\right)$	$\lambda^{(1)}_{X_{17}}\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda_{X_{18}}}$

Quantity: $\lambda_{X_{18}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{18}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{18}}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 6\lambda_{X_9} + 8\lambda_{X_{18}})$
T2	$+\lambda_{X_{18}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 6\lambda_{X_9} + 8\lambda_{X_{18}})$	$+\lambda_{X_{18}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 10\lambda_{X_9} + 8\lambda_{X_{18}})$	—	$-2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 10\lambda_{X_9} + 8\lambda_{X_{18}})$	—	$-2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 2 of 5

Column	Representative term structure
T1	$+12\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$
T2	$+24\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$
T3	$-\frac{9{g_2}^2\lambda_{X_{18}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+12\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+24\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$	$-\frac{9{g_2}^2\lambda_{X_{18}}}{2}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+40\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{18}}}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+40\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{18}}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 3 of 5

Column	Representative term structure
T1	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$
T2	$+6\lambda_1\lambda_{X_{18}}$
T3	$+6\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$	$+6\lambda_1\lambda_{X_{18}}$	$+6\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$	$+6\lambda_1\lambda_{X_{18}}$	$+2\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$	$+6\lambda_1\lambda_{X_{18}}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_{S_1}^{(1)}\lambda_{X_{18}}$
T2	$+\lambda_{X_{18}}(2T_{\nu}^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$
T3	$-4\lambda_{S_1}^{(A)}\lambda_{X_{18}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	$+\lambda_{X_{18}}(2T_{\nu}^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$	$-4\lambda_{S_1}^{(A)}\lambda_{X_{18}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	$+\lambda_{X_{18}}(2T_{\nu}^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right) + 2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right) + 6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right))$	$-4\lambda_{S_1}^{(A)}\lambda_{X_{18}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\mu_{\mathrm{T3GeneralS1S2}}}$

Quantity: $\mu_{\mathrm{T3GeneralS1S2}}$

$$16\pi^2\,\mu\frac{d\mu_{\mathrm{T3GeneralS1S2}}}{d\mu} = \beta^{(1)}_{\mu_{\mathrm{T3GeneralS1S2}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mu_{\mathrm{T3GeneralS1S2}}(T_\nu^{(1)} + T_\nu^{(2)})$
T2	$+12\lambda_9\,(\mu_{\mathrm{T3GeneralS1S2}})^*$
T3	$-\frac{9{g_2}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$\mu_{\mathrm{T3GeneralS1S2}}(T_\nu^{(1)} + T_\nu^{(2)})$	$+12\lambda_9\,(\mu_{\mathrm{T3GeneralS1S2}})^*$	$-\frac{9{g_2}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	$+12\lambda_9\,(\mu_{\mathrm{T3GeneralS1S2}})^*$	$-\frac{9{g_2}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1S2}}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3{g_1}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$
T2	$+2\lambda_{12}^{(1)}\mu_{\mathrm{T3GeneralS1S2}}$
T3	$-\frac{3\lambda_{12}^{(A)}\mu_{\mathrm{T3GeneralS1S2}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-\frac{3{g_1}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$	$+2\lambda_{12}^{(1)}\mu_{\mathrm{T3GeneralS1S2}}$	$-\frac{3\lambda_{12}^{(A)}\mu_{\mathrm{T3GeneralS1S2}}}{2}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-\frac{3{g_1}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$	$+2\lambda_{12}^{(1)}\mu_{\mathrm{T3GeneralS1S2}}$	$-\frac{3\lambda_{12}^{(A)}\mu_{\mathrm{T3GeneralS1S2}}}{2}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1S2}}$

Terms 3 of 3

Column	Representative term structure
T1	$+6\lambda_{10}m_2^2$
T2	$+6\lambda_{11}m_1^2$
T3	$\mu_{\mathrm{T3GeneralS1S2}}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+6\lambda_{10}m_2^2$	$+6\lambda_{11}m_1^2$	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+6\lambda_{10}m_2^2$	$+6\lambda_{11}m_1^2$	$\mu_{\mathrm{T3GeneralS1S2}}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\mu_{\text{T3GeneralS1Sq}}}$

Quantity: $\mu_{\text{T3GeneralS1Sq}}$

$$16\pi^2\,\mu\frac{d\mu_{\text{T3GeneralS1Sq}}}{d\mu} = \beta^{(1)}_{\mu_{\text{T3GeneralS1Sq}}}$$

Terms 1 of 4

Column	Representative term structure
T1	$m_1^2(4\,\mathrm{Tr}\left(y_1^{\text{conj}}\ (y_1)^\dagger\right) + 6\lambda_{X_9})$
T2	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\text{conj}}\right)^T\ (y_1)^*\right)$
T3	$+\mu_{\text{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$m_1^2(4\,\mathrm{Tr}\left(y_1^{\text{conj}}\ (y_1)^\dagger\right) + 6\lambda_{X_9})$	$-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\text{conj}}\right)^T\ (y_1)^*\right)$	$+\mu_{\text{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\right))$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$m_1^2(2\,\mathrm{Tr}\left(y_1^{\text{conj}}\ (y_1)^\dagger\right) + 10\lambda_{X_9})$	$-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\text{conj}}\right)^T\ (y_1)^*\right)$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$m_1^2(2\,\mathrm{Tr}\left(y_1^{\text{conj}}\ (y_1)^\dagger\right) + 10\lambda_{X_9})$	$-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ \left(y_1^{\text{conj}}\right)^T\ (y_1)^*\right)$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1Sq}}$

Terms 2 of 4

Column	Representative term structure
T1	$+24\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
T2	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$
T3	$+6\lambda^{(1)}_{X_5}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+24\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	$+6\lambda^{(1)}_{X_5}m_2^2$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+40\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	$+2\lambda^{(1)}_{X_5}m_2^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+40\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\text{T3GeneralS1Sq}}$

Terms 3 of 4

Column	Representative term structure
T1	$+\mu_{\text{T3GeneralS1Sq}}(2T_{\nu}^{(1)} + 2\,\text{Tr}\left(y_1^{\text{conj}}\,\left(y_1^{\text{conj}}\right)^{\dagger}\right))$
T2	$-12{g_2}^2\mu_{\text{T3GeneralS1Sq}}$
T3	$-4\lambda_{S_1}^{(A)}\mu_{\text{T3GeneralS1Sq}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\mu_{\text{T3GeneralS1Sq}}(2T_{\nu}^{(1)} + 2\,\text{Tr}\left(y_1^{\text{conj}}\,\left(y_1^{\text{conj}}\right)^{\dagger}\right))$	$-12{g_2}^2\mu_{\text{T3GeneralS1Sq}}$	$-4\lambda_{S_1}^{(A)}\mu_{\text{T3GeneralS1Sq}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\mu_{\text{T3GeneralS1Sq}}(2T_{\nu}^{(1)} + 2\,\text{Tr}\left(y_1^{\text{conj}}\,\left(y_1^{\text{conj}}\right)^{\dagger}\right))$	$-12{g_2}^2\mu_{\text{T3GeneralS1Sq}}$	$-4\lambda_{S_1}^{(A)}\mu_{\text{T3GeneralS1Sq}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1Sq}}$

Terms 4 of 4

Column	Representative term structure
T1	$+m_2^2(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+m_2^2(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	—
E	-4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\mu_{\text{T3GeneralS2Sq}}}$

Quantity: $\mu_{\text{T3GeneralS2Sq}}$

$$16\pi^2\,\mu\frac{d\mu_{\text{T3GeneralS2Sq}}}{d\mu} = \beta^{(1)}_{\mu_{\text{T3GeneralS2Sq}}}$$

Terms 1 of 4

Column	Representative term structure
T1	$m_2^2(2\,\text{Tr}\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\right) + 10\lambda_{X_2})$
T2	$-4\,\text{Tr}\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\ (M_F)^\dagger\right)$
T3	$+\mu_{\text{T3GeneralS2Sq}}(2T_\nu^{(2)} + 2\,\text{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$m_2^2(2\,\text{Tr}\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\right) + 10\lambda_{X_2})$	$-4\,\text{Tr}\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\ (M_F)^\dagger\right)$	$+\mu_{\text{T3GeneralS2Sq}}(2T_\nu^{(2)} + 2\,\text{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right))$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$m_2^2(4\,\text{Tr}\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\right) + 6\lambda_{X_2})$	$-8\,\text{Tr}\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\ (M_F)^\dagger\right)$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$m_2^2(2\,\text{Tr}\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\right) + 10\lambda_{X_2})$	$-4\,\text{Tr}\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\ (M_F)^\dagger\right)$	$+\mu_{\text{T3GeneralS2Sq}}(2T_\nu^{(2)} + 2\,\text{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right))$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS2Sq}}$

Terms 2 of 4

Column	Representative term structure
T1	$+40\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
T2	$-12{g_2}^2\mu_{\mathrm{T3GeneralS2Sq}}$
T3	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+40\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-12{g_2}^2\mu_{\mathrm{T3GeneralS2Sq}}$	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+24\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	—	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+40\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-12{g_2}^2\mu_{\mathrm{T3GeneralS2Sq}}$	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS2Sq}}$

Terms 3 of 4

Column	Representative term structure
T1	$-4\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$
T2	$+2\lambda_{X_7}^{(1)}m_1^2$
T3	$+\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	$+2\lambda_{X_7}^{(1)}m_1^2$	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+6\lambda_{X_7}^{(1)}m_1^2$	$+\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\left(y_2^{\mathrm{conj}}\right)^\dagger\right))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS2Sq}}$

Terms 4 of 4

Column	Representative term structure
T1	$+m_1^2(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	$+m_1^2(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$
E	-4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{y_1^{\mathrm{conj}}}$

Quantity: y_1^{conj}

$$16\pi^2\,\mu\frac{dy_1^{\mathrm{conj}}}{d\mu} = \beta^{(1)}_{y_1^{\mathrm{conj}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)(y_1)_{ij}$
T2	$+2T_\nu^{(1)}\left(y_1^{\mathrm{conj}}\right)_{ij}$
T3	$+2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)\left(y_1^{\mathrm{conj}}\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)(y_1)_{ij}$	$+2T_\nu^{(1)}\left(y_1^{\mathrm{conj}}\right)_{ij}$	$+2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,\left(y_1^{\mathrm{conj}}\right)^\dagger\right)\left(y_1^{\mathrm{conj}}\right)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)(y_1)_{ij}$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\,(y_1)^\dagger\right)(y_1)_{ij}$	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_1^{conj}

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{1}{4}(10\left(y_1\ (y_1)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 10\left(y_1^{\text{conj}}\ (y_1)^\dagger\ y_1\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 4\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij})$
T2	$-\frac{9}{2}{g_2}^2\left(y_1^{\text{conj}}\right)_{ij}$
T3	$-\frac{3}{2}{g_1}^2\left(y_1^{\text{conj}}\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{1}{4}(10\left(y_1\ (y_1)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 10\left(y_1^{\text{conj}}\ (y_1)^\dagger\ y_1\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 4\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij})$	$-\frac{9}{2}{g_2}^2\left(y_1^{\text{conj}}\right)_{ij}$	$-\frac{3}{2}{g_1}^2\left(y_1^{\text{conj}}\right)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\left(y_1^{\text{conj}}\right)_{ij}$	$-\frac{3}{2}{g_1}^2\left(y_1^{\text{conj}}\right)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\frac{1}{12}(29\left(y_1\ (y_1)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 29\left(y_1^{\text{conj}}\ (y_1)^\dagger\ y_1\right)_{ij} + 9\left(y_2\ (y_2)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_1^{\text{conj}}\right)_{ij} + 18\left(y_1^{\text{conj}}\ \left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij})$	$-\frac{9}{2}{g_2}^2\left(y_1^{\text{conj}}\right)_{ij}$	$-\frac{3}{2}{g_1}^2\left(y_1^{\text{conj}}\right)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_1^{conj}

Terms 3 of 3

Column	Representative term structure
T1	$+T_\nu^{(1)}\left(y_1^{\text{conj}}\right)_{ij}$
T2	$+\operatorname{Tr}\left(y_1^{\text{conj}}\left(y_1^{\text{conj}}\right)^\dagger\right)\left(y_1^{\text{conj}}\right)_{ij}$
T3	$+\frac{1}{12}(29\left(y_1\ (y_1)^\dagger\ y_1^{\text{conj}}\right)_{ij}+29\left(y_1^{\text{conj}}\ (y_1)^\dagger\ y_1\right)_{ij}+6(\left(y_2\ (y_2)^\dagger\ y_1^{\text{conj}}\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1^{\text{conj}}\right)_{ij}+3\left(y_1^{\text{conj}}\left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij}))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+T_\nu^{(1)}\left(y_1^{\text{conj}}\right)_{ij}$	$+\operatorname{Tr}\left(y_1^{\text{conj}}\left(y_1^{\text{conj}}\right)^\dagger\right)\left(y_1^{\text{conj}}\right)_{ij}$	$+\frac{1}{12}(29\left(y_1\ (y_1)^\dagger\ y_1^{\text{conj}}\right)_{ij}+29\left(y_1^{\text{conj}}\ (y_1)^\dagger\ y_1\right)_{ij}+6(\left(y_2\ (y_2)^\dagger\ y_1^{\text{conj}}\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1^{\text{conj}}\right)_{ij}+3\left(y_1^{\text{conj}}\left(y_1^{\text{conj}}\right)^\dagger\ y_1^{\text{conj}}\right)_{ij}))$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+T_\nu^{(1)}\left(y_1^{\text{conj}}\right)_{ij}$	$+\operatorname{Tr}\left(y_1^{\text{conj}}\left(y_1^{\text{conj}}\right)^\dagger\right)\left(y_1^{\text{conj}}\right)_{ij}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{y_2^{\text{conj}}}$

Quantity: y_2^{conj}

$$16\pi^2\,\mu\frac{dy_2^{\text{conj}}}{d\mu} = \beta^{(1)}_{y_2^{\text{conj}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$2\,\mathrm{Tr}\left(y_2^{\text{conj}}\ (y_2)^\dagger\right)(y_2)_{ij}$
T2	$+T_\nu^{(2)}\left(y_2^{\text{conj}}\right)_{ij}$
T3	$+\,\mathrm{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right)\left(y_2^{\text{conj}}\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$2\,\mathrm{Tr}\left(y_2^{\text{conj}}\ (y_2)^\dagger\right)(y_2)_{ij}$	$+T_\nu^{(2)}\left(y_2^{\text{conj}}\right)_{ij}$	$+\,\mathrm{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right)\left(y_2^{\text{conj}}\right)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$4\,\mathrm{Tr}\left(y_2^{\text{conj}}\ (y_2)^\dagger\right)(y_2)_{ij}$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$2\,\mathrm{Tr}\left(y_2^{\text{conj}}\ (y_2)^\dagger\right)(y_2)_{ij}$	$+T_\nu^{(2)}\left(y_2^{\text{conj}}\right)_{ij}$	$+\,\mathrm{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right)\left(y_2^{\text{conj}}\right)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_2^{conj}

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2\ (y_2)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2^{\text{conj}}\ (y_2)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 18\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$
T2	$-\frac{9}{2}{g_2}^2\left(y_2^{\text{conj}}\right)_{ij}$
T3	$-\frac{3}{2}{g_1}^2\left(y_2^{\text{conj}}\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+\frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2\ (y_2)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2^{\text{conj}}\ (y_2)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 18\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$	$-\frac{9}{2}{g_2}^2\left(y_2^{\text{conj}}\right)_{ij}$	$-\frac{3}{2}{g_1}^2\left(y_2^{\text{conj}}\right)_{ij}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$-\frac{9}{2}{g_2}^2\left(y_2^{\text{conj}}\right)_{ij}$	$-\frac{3}{2}{g_1}^2\left(y_2^{\text{conj}}\right)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+\frac{1}{12}(9\left(y_1\ (y_1)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2\ (y_2)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 29\left(y_2^{\text{conj}}\ (y_2)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2^{\text{conj}}\right)_{ij} + 18\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij})$	$-\frac{9}{2}{g_2}^2\left(y_2^{\text{conj}}\right)_{ij}$	$-\frac{3}{2}{g_1}^2\left(y_2^{\text{conj}}\right)_{ij}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: y_2^{conj}

Terms 3 of 3

Column	Representative term structure
T1	$+2T_\nu^{(2)}\left(y_2^{\text{conj}}\right)_{ij}$
T2	$+2\,\text{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right)\left(y_2^{\text{conj}}\right)_{ij}$
T3	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2^{\text{conj}}\right)_{ij}+2(5\left(y_2\ (y_2)^\dagger\ y_2^{\text{conj}}\right)_{ij}+5\left(y_2^{\text{conj}}\ (y_2)^\dagger\ y_2\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_2^{\text{conj}}\right)_{ij}+2\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij}))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+2T_\nu^{(2)}\left(y_2^{\text{conj}}\right)_{ij}$	$+2\,\text{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right)\left(y_2^{\text{conj}}\right)_{ij}$	$+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2^{\text{conj}}\right)_{ij}+2(5\left(y_2\ (y_2)^\dagger\ y_2^{\text{conj}}\right)_{ij}+5\left(y_2^{\text{conj}}\ (y_2)^\dagger\ y_2\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_2^{\text{conj}}\right)_{ij}+2\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2^{\text{conj}}\right)_{ij}))$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—